Estimating Historical Implied Volatility

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Abstract

Decades after key historical events in the late 19th and 20th centuries, the spirit of bygone eras - fear, uncertainty, euphoria - and the concerns of average investors are lost in hard data. Since 1986, VXO and VIX served as the preeminent gauge of expected equity market volatility and investor sentiment, but are not reported prior due to lack of options price data. I answer the question of what implied volatility, a market-based measure of the spirit of the times, likely was during seminal historical periods, like the Panic of 1907 or the Cuban Missile Crisis of 1962. Leveraging rich historical textual news coverage from the Wall Street Journal, I use a novel transfer learning approach to produce an empirically grounded estimate of historical implied volatility spanning July 1889 to December 2009. This times series augments the historical record and has a host of applications in empirical asset pricing.

1 Introduction

On October 30th, 1929, a headline of the Wall Street Journal read "STOCKS STEADY AFTER DECLINE: Bankers State Support Continues-Spokesman Expresses View Hysteria is Passing: RECORD TRADING VOLUME". The Dow Jones Industrial average fell 13% on Black Monday and 12% on Black Tuesday, the two prior days, but the sell-off was purportedly a passing hysteria. It certainly did not portend a decade-long Great Depression. With the benefit of hard data, like stock returns, employment, and industrial production figures, it is obvious decades later that this event augured ill for the global economy. However, no such data captures the spirit of the times, of what people in the moment perceived. Historians rely on primary source evidence like written accounts to capture the ideas, attitudes, and feelings of past eras. I set out to empirically measure this spirit of the times with a historical estimate of stock index implied volatility, generated from the pattern of co-movement between option implied volatility and textual news coverage from the Wall Street Journal.

Implied volatility is the volatility of the underlying security that equates the observed market options price to the Black-Scholes options price. Implied volatility is a forward-looking measure, reflecting investors' expectations about the volatility of the underlying security. Since 1986, the Chicago Board Options Exchange has produced daily measures of implied volatility[1]. Returns data is plentiful prior to 1986, but options price data is not. An estimate of implied volatility prior to 1986 uniquely augments macroeconomic history with countless applications to empirical asset pricing. Realized volatility of a security or index is a measure of return fluctuations that have occurred over some period. In practice, realized volatility measures exhibit time-varying variance, fat-tailedness, and a right skew

[1]VXO debuted in 1986 for the S&P100 index and was replaced by VIX in 1993. VIX initially measured the market's expectation of S&P100 index 30-day volatility and was updated in 2003 to measure expected volatility of the S&P500 U.S. equity index by aggregating weighted prices of call and put equity options.

(Opschoor and Lucas 2023). Historical realized volatility data captures periods of market turbulence, often coinciding with extreme events like the terrorist attacks of September 11th, 2001. Daily index values for the Dow Jones Industrial Average are available from May 26th, 1896, and daily S&P500 values from 1926. I use measures of realized volatility for robustness and a novel use of transfer learning to estimate historical implied volatility.

I use the digitized body of historical daily Wall Street Journal articles from July 7th, 1889, to December 31st, 2009 for text data. The Journal has covered key economic and financial events, politics, wars, domestic unrest, natural disasters, technological and scientific advancements, and cultural and social movements for decades. Each front-page is a snapshot of history; over many years, the news coverage reflects the changing composition of events relevant to investors and financial markets. The dataset has a longer span than many macroeconomic time series and enables this paper. With basic natural language processing techniques, I transform this body of text into numerical arrays used as inputs to high-dimensional regression models that relate a month's worth of WSJ text to month-end implied volatility.

Before conducting detailed analyses, I first examine whether text features relate to implied volatility by regressing differences in implied volatility levels between any two months on the textual similarity of those months' news content in the Journal. Using the cosine similarity of text to proxy the similarity of news content, I identify a statistically significant decreasing relationship between textual similarity and difference in level of implied volatility over the January 1986 to Decmber 2009. Months with more similar news coverage tend to have similar levels of implied volatility, and this intuitive result is robust to different specifications. I then use latent Dirichlet allocation, an unsupervised topic model, to extract latent topics from the post-1986 sample of Journal articles and identify well-formed and temporally delineated topics that reflect key events from the his-

torical record. Text data from the Wall Street Journal does in fact capture the important economic and political developments of specific time periods.

I experiment with combinations of a range of models, text pre-processing and vectorization techniques, and training methodologies to identify the best approach to estimating historical implied volatility using only data between January 1986 December 2009 on the co-movement between the front-page of the Journal and options implied volatility[2]. A gradient boosting regressor achieves peak predictive performance of 4.96 RMSE 0.48 R^2 when trained on an 80% split of count-vectorized text with a vocabulary of 342,155 text features. Predicting implied volatility over the July 1889 to December 1985 period using this model however yields a series seemingly indifferent to spikes in historical realized volatility and at odds with stylized facts regarding the variance risk premium.

With a long out-of-sample prediction period, this task requires a meaningful empirical test. The variance risk premium is measured as the difference between option implied and realized variance, and reflects the nature of time-varying volatility and investors' risk aversion. From January 1991 to December 2021, S&P500 option and realized variance have a 0.78 correlation and the premium averages 15.09, but is negative during extreme crises. I argue that no conventionally trained model[3] can produce a historical implied volatility series that coheres with these facts, owing to insufficient data and proclivity to overfitting and the fundamental differences in the historical records. There are no post-1986 historical analogues to World War I, World War II, the Spanish Flu of 1918, or the Cuban Missile Crisis of 1962, among countless other historical events. Transfer learning addresses each technical problem and is uniquely posed to leverage the rich history of the

[2]Fixed vocabulary sizes span 468,091 to a mere 5,000 unique words or phrases, models include gradient boosting, support vector, penalized linear, partial least squares, principal components, and multi-layer perceptron regression. I vectorize with count and term frequency-inverse document frequency vectorization, and use either a randomized or temporal split of the sample into training and testing subsamples

[3]Conventional means trained exclusively on a subset of the January 1986 to December 2009 period.

training and prediction periods and generate an empirically grounded, rigorously derived, and historically accurate estimate of implied volatility.

Transfer learning is a machine learning technique that involves taking a model trained in one domain and redeploying it to another. This paper's key insight is that a neural network can learn the relationships between textual news coverage and Dow Jones Industrial Average realized volatility over the period of May 1896 to December 1985, accounting for decades of history, and can be modified to learn the distinction between realized and implied volatility using data from the January 1986 to December 2009 period. I outline a procedure that involves freezing the network's early hidden layers to preserve learned feature representations from the realized volatility period. I document increased implied volatility predictive performance of 0.72 R^2 and an RMSE of 1.50 despite using only 5,000 text features, a remarkable combination of data reduction and performance uplift. Deep historical knowledge gives a model more predictive power for implied volatility because the model has been exposed to events like the Panics of 1893 and 1907, two World Wars, the Great Depression, and the inflationary period of the 1970s, and their relationship with realized volatility. With a longer history, less text data about that history is needed.

Transfer learning estimated historical implied volatility co-moves with realized volatility but is smoother and higher on average. During extreme crises, implied volatility is lower than realized volatility, matching the post-1990 variance risk premium's pattern. It responds to exogenous shocks like attacks and assassinations and the levels in seminal periods accord with the historical record. Estimated implied volatility is lower than realized volatility during the Crash of 1929, reflecting the sentiment - expressed on the front-page of the Journal - that despite the huge liquidation volume, markets would soon stabilize and the prosperity of the Roaring Twenties would continue. The estimated series captures the spirit of the times since July 1889 and offers unique insights into financial history.

2 Literature Review

This paper is motivated by Manela and Moreira's (2017) attempt to address the lack of VXO data prior to 1986 by constructing a text-based measure of uncertainty. They create a corpus of front-page articles from the Wall Street Journal stretching back to 1889, under the assumption that the business press coverage of particular topics is a proxy for investors' concerns about such topics. They train a support vector regression to identify co-movement between observed VXO and the front-page of the Wall Street Journal, evaluate the model's performance on a test period, and then predict implied volatility using WSJ articles prior to 1986. They label this predicted series NVIX. Aside from the historical implausibility of their estimates[4], I document several flaws with their work including over-stated performance metrics and irreproducible results[5]. I rely on their published data to access the set of front-page WSJ articles in processed form, meaning I obtain their vocabulary and counts of words per month but not the raw text. This forecloses more sophisticated natural language processing techniques and limits the text data to December 2009, but I show strong results despite these limitations. No papers attempt to improve on their NVIX estimate and I couldn't find any significant papers that use NVIX. The series has been used for return and volatility predictability tests in the Asia-Pacific (Su et al 2018), but I couldn't find other applications.

Uncertainty shocks are studied extensively in the asset pricing literature. A key tenet in financial economics is that asset returns are driven in part by unanticipated fundamental information. Econometric approaches include the development a structural model to simulate uncertainty shocks and their macroeconomic effects (Bloom 2009) and shock-restricted structural vector autogressions which constrain the sign, magnitude, and corre-

[4]See Appendix 11.10 for a historical analysis of the trend of their series.

[5]See Appendix 11.8 for discrepancies in their reported measures and 11.9 for return predictability.

lation of shocks during unusual historical episodes (Ludvigson et al 2021). The relationship between surges in uncertainty and fluctuations in asset prices and business cycles is well documented, but more recent works study theories of belief formation in which agents revise estimates of disaster risks based on previously unobserved tail events (Orlik and Veldkampf 2022). I carryover many of the assumptions in these papers regarding what constitutes an uncertainty shock, the transmission mechanism between uncertainty and market activity, and the importance of uncertainty shocks to asset pricing.

Several papers use text-based machine learning models. I use the basic theoretical framework of document-term matrices and text regression as characterized in Gentzkow et al (2019). Garcia (2013) uses negative and positive words in New York Times articles to study the effect of sentiment on asset prices, but I found this deterministic approach has little predictive power and generalizes poorly to earlier Journal articles. Similarly, Baker et al (2016) constructs an economic policy uncertainty index from counts of key words in news articles. This index would be useful in robustness tests but the lack of raw WSJ text data makes implementation difficult. I view the rationale provided in this paper as bolstering my use of text regressions. Boudoukh et al (2018) uses text analysis to identify fundamental information in the news that accounts for overnight idiosyncratic volatility. I keep the type and tone of financially relevant news cited in this paper in mind when pruning words and phrases from the Journal articles and when conducting unsupervised topic modelling. Loughran and McDonald (2011) similarly develop a negative word list for tone assessment of financial texts. I found the performance of these word list and sentiment based approaches insufficient to warrant their usage. Ke et al (2019) examines return forecasts and idiosyncratic returns using text data, and incorporates topic modelling to assign prediction weights to specific words. Their list of terms is informative, as is their documentation of the assimilation of news information into asset prices. Kelly et

al (2021) develops a high-dimensional text selection model and shows that WSJ text data improves forecasting of macroeconomic indicators. I also surveyed the extensive computer science and machine learning literature on text regression. Kogan et al (2009) use a support vector regression on financial reports to predict stock return volatility. Support vector regressions are hardly novel; papers on support vector regressions date back to 1997. Other approaches to text regression include elastic net (Joshi et al 2010). Bitvai and Cohn (2015) develop a deep convolutional neural network for text regression. Deep CNNs are considered state of the art but involve more sophisticated text representations and complicated neural architectures. My early attempts to duplicate such architectures revealed their success is contingent on exhaustive calibration to a specific dataset.

Research on stochastic volatility and the variance risk premium includes studies of variance swap rates (Wu and Carr 2007). I also examined research on volatility forecasting. Bekaert and Hoerova (2014) evaluate state-of-the-art volatility forecasting time series methods and show that VIX has predictive power for returns, economic activity, and financial instability. I found their methods difficult to repurpose for back-predicting historical values. For basic exposition on the variance risk premium, I refer to Zhou (2018). Transfer learning has been used to improve cross-sectional model generalisability, reduce over-fitting, and improve performance for securities with limited samples (Poh et al 2023). This paper confirmed my intuition that transfer learning is appropriate given the relative scarcity of implied volatility data. A critical application is the development of FinBERT (Huang et al 2022), a large language model adapted to the finance domain that can classify the sentiment of financial text. Given the varied nature of Journal articles, with many detailing non-financial events, I did not pursue sentiment classification any further. Transfer learning is rare in finance resarch. To the best of my knowledge, this paper is the first to use transfer learning to estimate historical implied volatility.

3 Data

3.1 WSJ Corpus

Charles Dow, Edward Jones, and Charles Bergstresser published the first edition of the Wall Street Journal on July 7th, 1889 (Murray 2019). News coverage at the time was often sensationalized and unreliable, and at times factually incorrect, a phenomena known as yellow journalism. At at time when William Randolph Hearst's New York Journal and Joseph Pulitzer's New York World dueled for circulation, the WSJ provided independent and factual reporting on financial, business, and economic news. Since 1889, the Journal has been printed continuously and has covered critical events from around the world, including the formation and dissolution of major corporations and industries, scientific and technological advances, military conflicts and terrorist attacks, natural disasters, pandemics and contagions, political and social movements, and financial crises.

Digitized copies of the front-page of every daily WSJ are available dating back to July 1889, constituting a body of news coverage unrivaled in its historical depth and richness. Event coverage reflects the publisher's perception of what is impactful and pertinent to readers' interests. This feedback mechanism has been calibrated over decades of publication, thus I contend that business press is a stable proxy for the information set and concerns of investors. Any individual Journal article chronicles discrete events, but the gestalt is a written record of the key events, concerns, sentiments, and views of bygone eras. A headline from December 8th, 1941, one day after Pearl Harbor, uniquely captures the astonishment of the nation and the precariousness of the moment. The variation in headlines thereafter captures growing uncertainty over the depth of US involvement in World War II. Manela and Moreira (2017) collect the title and abstract of all front-page WSJ articles from July 1889 to December 2009, perform manual corrections to improve

quality, and aggregate daily text to a monthly frequency. The result is large body of text per month for 1,439 months. I utilize their data because I couldn't get access to the original digitized WSJ corpus[6].

Despite these limitations, I am able to alter some of their text pre-processing steps to test the effects of data reduction and perform topic modelling. For each month's worth of text, I count the number of words in that month and the number of unique words. Total word count is unsurprisingly stable over time, given the physical constraints of a newspaper's front-page, whereas the number of unique words per month varies considerably.

Table 1: WSJ Corpus Summary Statistics

	Count of Unique Words	Count of Total Words
Count	1439	1439
Mean	7081.64	885228.40
Std.	4705.87	40681.12
Min.	2111	807211
p25	3646	845387.50
p50	5787	879357
p75	8285.50	926073
Max.	24937	955817

Count of unique and total words is per monthly body of text. Daily articles are combined per month.

Common financial and economic words include tax, rates, money, stock, market, industry, bank, credit, and plan. Words pertaining to politics and conflict include US, Washington, war, violence, and military. There are also words pertinent to disasters, such as disease, fire, storm, and aid pertain to disasters. Names of individuals, locations, and institutions feature prominently. See Table 2 for a sample of Wall Street Journal headlines and abstracts featured originally in an article commemorating the history of the Journal.

[6]Lack of data access limits my efforts in several regards. First, I cannot extend the analysis past December 2009. Second, I cannot disaggregate a month's worth of text into finer frequencies, such as daily or weekly. Third, I cannot reconstruct the actual text as it appears in the WSJ owing to the orderless property of a document-term matrix, precluding more sophisticated textual analysis techniques.

Table 2: Sample of WSJ Headlines

Date	Headline	Abstract
May 16, 1911	STANDARD OIL CO. DISSOLUTION ORDERED BY SUPREME COURT.	Only Change Relieves the Corporation From Immediate Restraint in Interstate Commerce and Extends From Thirty Days to Six Months the Time in Which to Dissolve and Qualify as a Lawful Body—Opinion By Chief Justice White—Sherman Act is Violated
August 5, 1914	GERMANY WARS ON FRANCE; ENGLAND STANDS BY BELGIUM	100,000 German Troops March to Invade France—Germany Appeals to Italy to Rejoin the Alliance—Switzerland Completes Mobilization—Skirmishing Along Prussian Frontier—English War Party in Power
June 14, 1930	SENATE PASSES TARIFF BILL 44-42	Measure Goes To House Today With Favorable Action Practically Assured
April 20, 1933	Says U.S. Off Gold	Secretary of Treasury Woodin Admits Official Abandonment Of Standard
December 8, 1941	Japan Attacks Hawaii, Damage and Loss of Life Reported Heavy; Japan Declares War on U.S., England	ROOSEVELT TO ADDRESS CONGRESS TODAY TOKYO ENVOY DELIVERS REPLY TO U.S.; HULL: "INFAMOUS," FILLED WITH LIES REDS CLAIM GAINS ON ALL FRONTS BRITISH, AXIS TANKS IN MAJOR BATTLE
May 9, 1945	Germany's Surrender	V-E Day Starts a Slow, Confused Unwinding of American War Industry Government Strives to Hold Its Price and Wage Controls, but Success Is Dubious; Arms Cutback Estimates Vary
February 3, 1956	GM 1955 Profits Topped $ Billion For 1st Time on $12.4 Billion Sales	CURTICE SEES NET FOR 1956 "NOT FAR OFF" FROM THIS YEAR'S RECORD FIGURE
November 15, 1972	The Dow Tops 1000, but That Tells Little About Past or Future	Widely Followed but Narrow Stock Average, 88 Years Old, Has Had Its Ups and Downs
October 17, 1973	OIL EMBARGO	'Cold War': Reduced Arab Oil Flow Promises to Make Life Difficult for Americans. Some Ways to Ease Hardship Seen; Firms Build Stocks; But Mideast Remains Vital. Laird's Advice: Buy Sweater
June 20, 1979	High-Priced Cure	Fight to Curb Inflation, Experts Say, Will Take Years and Risk a Slump If Joblessness Starts Rising, Will Washington Panic? Woes of Paring Programs
November 20, 1987	TERRIBLE TUESDAY: OCT. 20, 1987	How the Stock Market Almost Disintegrated a Day After the Crash
September 12, 2001	A DAY OF TERROR	Eye of the Storm: A Nightmare of Falling Bodies, Acrid Smoke and Heroism; 'It's Coming Down! Run!'
September 15, 2008	Crisis on Wall Street as Lehman Totters, Merrill Is Sold, AIG Seeks to Raise Cash	Fed Will Expand Its Lending Arsenal In a Bid to Calm Markets; Moves Cap A Momentous Weekend for American Finance

3.2 Implied Volatility

I compute summary statistics for VXO during the train and test periods using month-end Chicago Board Options Exchange VXO[7]. VXO is on on average higher and more volatile during the training period of 1996.01-2009.12 than the test period of 1986.01 - 1995.12. Manela and Moreira define the specified training and testing periods.

Table 3: VXO Summary Statistics

	1986.01-2009.12	1996.01-2009.12	1986.01-1995.12
Count	288	168	120
Mean	21.37	23.24	18.75
Std.	8.64	9.17	7.07
Min.	9.82	10.14	9.82
p25	15.02	16.73	13.75
p50	19.87	23.01	18.11
p75	24.91	26.54	21.35
Max.	61.41	61.38	61.41
Note:	CBOE month-end VXO (monthly freq.).		

I perform t-tests to compare the means, assuming equal variances and unequal variances, and for both reject the null hypothesis of equal means. I also perform a Kolmogorov-Smirnov test and reject the null hypothesis that both samples are drawn from the same distribution. For all tests, the p-value associated with the test statistic is near zero.

Table 4: Statistical Tests

Test	Null	Alternative	Test Statistic	P-Value
T-Test	$\mu_1 = \mu_2$	$\mu_1 \neq \mu_2$	4.66	0.00
Kolmogorov-Smirnov	F_1,$F_2 \in F$	F_1,$F_2 \notin F$	0.353	0.00

The sample differences highlight the challenge of training models with VXO as a dependent variable. Aberrationally high or low VXO in the training sample can cause a model to overfit and predict similar levels for the test sample. I discuss this further in Section 5.4.

[7]For five months, VXO on the last day in the month is unavailable I use VXO from the last available day in that month

3.3 Realized Volatility

I employ two measures of realized volatility for robustness, estimates of the variance risk premium, and transfer learning. The primary measure is the monthly Dow Jones Industrial Average volatility, where realized variance is the non-overlapping sum of squared log returns for month index t, n observations per month, and daily index value S_{t+i}:

$$\mathrm{RV}_{t,t+1} = \sum_{i=1}^{n} \ln\left(\frac{S_{t+i}}{S_{t+i-1}}\right)^2 \tag{1}$$

All realized volatility data is sourced from CRSP. Daily DJI data from May 1896 is available, except for a short period around the outbreak of WWI[8]. Research on volatility often uses the sum of squared log intra-day returns at narrow intervals, such as every 5 minutes. I find that my measure using historical daily values resembles and has a 0.98 correlation with estimated volatility from high frequency data[9]. A second measure uses the standard deviation of a month's worth of daily S&P500 returns.

Table 5: Realized Volatility Summary Statistics

	SP500	DJIA
	1926.07-2009.12	1896.05-2009.12
Count	1001	1360
Mean	13.81	15.52
Std.	9.58	9.71
Min.	2.76	4.32
p25	8.27	9.81
p50	10.82	12.99
p75	15.52	17.58
Max.	79.23	99.89

SP500 and DJIA monthly volatility, computed from daily returns and annualized by a factor of $\sqrt{12}$.

Recession indicators are sourced from the NBER time series of US business cycle expansions and contractions, available on FRED. Recessionary months are shaded in grey.

[8]See Appendix 11.1 for a plot.

[9]See Appendix 11.2 for a comparison of DJIA (daily returns) and SP500 volatility (intra-day returns).

4 Preliminary Analysis

Under the hypothesis that text features are a sufficient statistic for the continuous variable represented by those features, two months of news with similar text should have similar levels of implied volatility. If a month's worth of news coverage fully encapsulates an investor's concerns and drives forward-looking uncertainty measures, then the differences between two months' texts should explain variation in levels of implied volatility. To test the plausibility of using text features to predict implied volatility, I regress the differences in implied volatility for any two months on the cosine similarity of the text associated with those months. Define the cosine similarity of two fixed-length vectors as:

$$\cos(\theta_{ij}) = \frac{\vec{x}_i \cdot \vec{x}_j}{||x_i||||x_j||} \tag{2}$$

for text vectors $\vec{x}_j$ and $\vec{x}_i$[10]. As noted, a single document can be represented as a fixed-length vector using the bag-of-words representation. For each month i, I transform i's news text into a fixed-length sparse vector of dimension equal to the total vocabulary size, and identify the month j ($j \neq i$) with the highest cosine similarity to i, forming a pair (i,j)[11]. I then regress the absolute difference in implied volatility on log cosine similarity[12] for each pair of months and test the model with and without a constant[13]:

$$|\text{VXO}_i - \text{VXO}_j| = \beta_0 + \beta_1 \cdot \ln(cos(\theta_{ij})) + \epsilon_{ij} \tag{3}$$

[10]Two identical vectors have an angle θ between them of 0 and thus cosine similarity of 1.

[11]The closest month for 1986.01, 1986.02, 1986.03 is 1995.06, for 1986.04 it is 1994.09, 1986.05 it's 1997.07, and so on. For 2009.08-2009.12, it is 2009.07.

[12]The logarithmic transformation is useful to interpret β_1. Although $-1 \leq \cos(\theta_{ij}) \leq 1$, token counts are non-negative so the lower bound here is 0. I increment by 0.01 to take a log: $\ln(0.01) \approx$ -4.61.

[13]As additional robustness checks, I test linear-linear and log-log models as well. To address the question of whether the regression is simply picking up sequential months that have similar coverage and volatility around large events, I plot the distance between closest months, rolling month by month from 1986.01 to 2009.12 and run regressions on the samples separated by 1993. In 1993, news coverage becomes highly persistent. See Appendix 11.5 for additional specifications and 11.6 for coverage persistence.

For two months with identical news coverage, $\ln(cos(\theta_{ij})) = 0$ and $\beta_0 = 0$, so I impose this restriction by running the regression without a constant. As cosine similarity between the texts increases, the absolute difference between their levels of implied volatility should decrease. The function $\ln(x)$ is negative and increasing over $0 \leq x \leq 1$, thus for a non-negative absolute difference in implied volatility I expect that $\beta_1 < 0$. For the model with an intercept, completely dissimilar text results in a difference of $\hat{\beta}_0 - 4.61\hat{\beta}_1$. I present

Table 6: Linear-Log Text Similarity Regression

	$\lvert \text{VXO}_i - \text{VXO}_j \rvert = \beta_0 + \beta_1 \cdot \ln(cos(\theta_{ij})) + \epsilon_{ij}$	
	No Intercept	Intercept
Log Cosine Similarity	-9.70***	-6.27***
	(-18.03)	(-6.14)
Constant		1.88***
		(3.91)
Observations	287	287
R^2	0.53	0.12
Adjusted R^2	0.53	0.11
Residual Std. Error	4.28	4.18
F Statistic	325.22***	37.68***
Note:	*p<0.1; **p<0.05; ***p<0.01. T-stats in parentheses.	

regression results above. Evidence in favor of the aforementioned hypothesis is a large R^2 and a statistically significant $\beta_1 < 0$ for the intercept-less model. Without the restriction, $\hat{\beta}_0$ is small but statistically different from zero and $\hat{\beta}_0 - 4.61\hat{\beta}_1 \approx 30.78$, which is one standard deviation above average VXO from 1986 to 2009. A one percent increase the cosine similarity between the text of two months decreases the absolute difference between their implied volatility levels by $6.27/100$[14].

A second test of the hypothesis that news text bears on expectations of uncertainty

[14] $\hat{\beta}_1 \ln(1.01) \approx 0.0624$

is whether pertinent topics, such as foreign relations, economic and financial news, or military conflicts and terrorism, can be extracted from news text. I use latent Dirichlet allocation, a generative probabalistic model in which text data arises from a generative process that includes latent variables, to extract a latent topic structure from the Jan. 1986 - Dec. 2009 WSJ. A document is a probability distribution over topics, represented by the vector θ_d, and a topic, represented by vector β_i, is a probability distribution of words. $\theta_{d,k}$ is the topic proportion for topic k in the dth document and z_d is the assignment of topics for document d. Words are drawn from a fixed vocabulary and represented by $w_{d,n}$, the nth word in the dth document (Blei 2012). The joint distribution is:

$$p(\beta_{1:K}, \theta_{1:D}, z_{1:D}, w_{1:D}) = \prod_{i=1}^{K} p(\beta_i) \prod_{d=1}^{D} p(\theta_d) \left(\prod_{n=1}^{N} p(z_{d,n}|\theta_d) p(w_{d,n}|\beta_{1:K}, z_{d,n}) \right) \tag{4}$$

and the conditional topic structure distribution:

$$p(\beta_{1:K}, \theta_{1:D}, z_{1:D}|w_{1:D}) = \frac{p(\beta_{1:K}, \theta_{1:D}, z_{1:D}, w_{1:D})}{p(w_{1:D})} \tag{5}$$

Gensim's implementation of LDA outputs the topic-word distributions and document-topic distributions per document. The former is a list of tokens per topic, where the user sets the number of topics, and the latter is a probability distribution. I examine the quality of the topic distributions and topic loadings per document. Each document is one month's worth of news articles, so the document-topic distribution represents the composition of that month's news in terms of latent topics. By examining the topic-word distribution, I get a sense of what topics prevail in each month, and whether news text captures actual events relevant to forward-looking investors. I present times series of $\hat{\theta}_d$, where d indexes one month of text from 1986.01 - 2009.12, and identify temporally delineated topics matching the historical record. Topic 1 spans the 2000 Bush-Gore Presidential and Florida recount, US involvement in Kosovo in 1999 under President Clinton, and the advent of the internet. Topic 3 spans accounting scandals and terrorism in the early

to mid-2000s. Topic 5 refers to Soviet and Japanese relations, inflation and recession concerns, and the Savings and Loan crisis of the 1980s. Topic 6 spans the Global Financial Crisis and the Obama administration in 2009. Topic 7 corresponds to the terrorist attacks on 9/11 and the beginning of the War on Terror in late 2001[15].

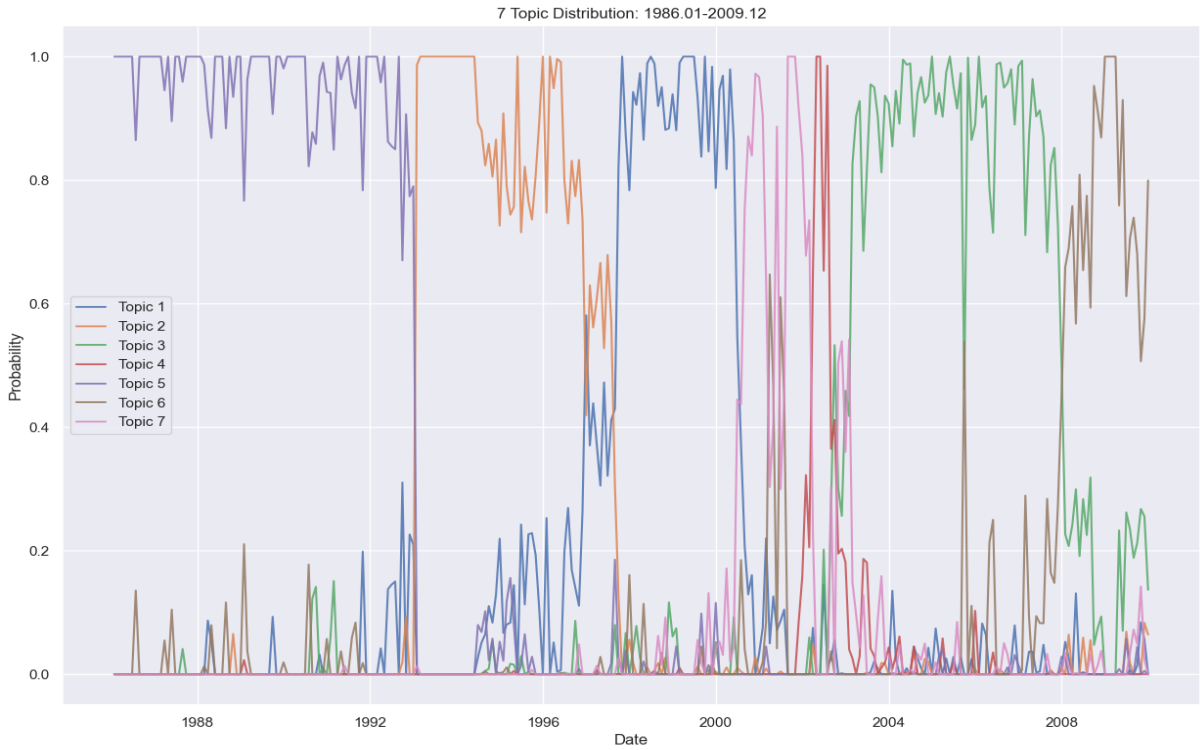


Figure 1: Monthly Topic Distributions, 1986.01-2009.12

Having established the relationship between textual distance and differences in implied volatility, I proceed with fitting text regression models, . Classification is similarly motivated by this exercise as it involves predicting volatility groups instead of levels.

[15]See Appendix 11.7 for VXO's sensitivity to the topic-distributions. Topic 7: attack, taliban, afghanistan, bush, bin, laden, terror, military, pakistan, troop, bomb, anthrax. Topic 1: gore, bush, vote, internet, campaign, gop, elect, clinton, kosovo, technology, florida. Topic 5:outlook, bush, recession, soviet, japanese, deficit, inflation, reagan, budget, west, loan, tokyo. Topic 6: fund, obama, credit, fed, mortgage, investor, global, crisis, share, loss, treasuries, bush, decline, insurance. Topic 3: enron, account, sec, worldcom, andersen, bush, aig, al, qaeda, israel.

5 Methodology

5.1 Document-Term Matrix

Text regression is the task of predicting a continuous value from text features. The theoretical framework for text regression is outlined in Gentzkow et al (2019):

1. Represent raw text D, the set of daily articles from the front-page of the WSJ spanning 1889 to 2009, as a numerical array C;

2. Map array C to predicted values of volatility $\widehat{\text{Vol}}$ from 1889 to 1986, a period in which options data and thus implied volatility are not available;

3. Use the predicted values $\widehat{\text{Vol}}$ of historical implied volatility in descriptive analysis[16].

C is a document-term matrix. A corpus is simply a body of text. Let the vocabulary of the WSJ corpus, V be the set of all unique tokens, which are words or phrases of length n[17]. After removing stopwords, which are especially common and uninformative words, and taking other pre-processing steps, such as removal of punctuation, numbers, lowercasing, and reducing tokens to their stemmed forms, the vocabulary represents the set of text features in the corpus[18]. By construction, the cardinality of the vocabulary, $|V|$, equals the number of columns in the document-term matrix. C is at most dimension 468,091 x 1439, corresponding to 468,091 text features and 1,439 months worth of text data between July 1889 and December 2009.

Document c_t is subset of the raw text D, and in theory can be constructed at any level of granularity; in this application, a document could be a day's worth of news, or a month's worth. Manela and Moreira chose a monthly frequency, and given the document-term matrix but not the raw text[19], I have no way of disaggregating a month's worth of

[16] I use descriptive analysis of the predicted values as a robustness check on the quality of the estimates.
[17] n is often 1, corresponding to unigrams, 2, corresponding to bigrams, or 3, corresponding to trigrams
[18] See Appendix 11.4 for a discussion of vocabulary reduction.
[19] See Appendix 11.3 for details on reconstructing the WSJ corpus.

text into daily or weekly documents. Regardless, representing a month's worth of news text as a document provides a reasonable balance between sample size, and thus statistical power, and sensitivity of the model's fit to aberrational text that may appear at the daily level. Hence each document c_t is one month's worth of front-page WSJ articles.

A document-term matrix employs a bag of words representation of text, in which text is an unordered set of words with only the frequency of a word's appearance, but not position, in the text retained (Jurafsky and Martin 2014). Element c_{tv} of the document-term matrix is a count of the number of times vocabulary element v appears in document c_t. Each document is a fixed-length row vector of dimension $1x|V|$. Each column c_v is a times series of frequencies of token v across documents $c_1....c_T$. This vector space model is the simplest in natural language processing but is highly tractable. Text features are covariates and the columns of the document-term matrix can be used as inputs for a host of high-dimensional statistical models. Text-data is inherently high-dimensional because each unique token or phrase is a feature. Restricting the vocabulary through conventional text pre-processing is thus a model-free form of dimensionality reduction and reducing computational cost. I document several measures to reduce the vocabulary and show that with just 5,000 unique tokens, comparable or superior out-of-sample predictive performance can be achieved.

An alternative to token counts is term frequency-inverse document frequency:

$$\begin{aligned} \mathrm{tf}_{ij} &\equiv \mathrm{c}_{ij} \\ \mathrm{idf}_j &\equiv \log\left(\frac{n}{\sum_i \mathbf{1}_{c_{ij}>0}}\right) \\ \mathrm{TFIDF}_{ij} &= \mathrm{tf}_{ij} \cdot \mathrm{idf}_j \end{aligned} \tag{6}$$

for text feature j in document i, count of occurrences c_{ij}, and the log of one over the share of documents containing j (Gentzkow 2019). TFIDF penalizes words that appear infrequently (low term frequency) in many documents (high inverse document frequency)

and assigns higher scores to words that appear frequently in a given document but sparsely throughout the collection.

I partition the document-term matrix C into train, test, and predict matrices, with a subset of the C^{train} matrix used to tune model hyperparameters. The train and test samples are constructed by selecting rows, either randomly or using a temporal split, from C. C^{predict} is the set of monthly documents from July 1889 to December 1985 over which implied volatility data is not available. Models are trained on C^{train}, evaluated out-of-sample on C^{test} by comparing $\widehat{\text{Vol}}_{\text{test}}$ to Vol_{test} with metrics including R^2 and mean squared error[20], and then fitted models are used to obtain $\widehat{\text{Vol}}_{\text{predict}}$ from C^{predict}.

Generative models estimate the conditional probability of text given an observed input, meaning a model of $p(c_t|\text{Vol}_t)$ instead of $p(\text{Vol}_t|c_t)$ (Gentzkow et al 2019). One may hypothesize that implied volatility itself drives news coverage, or that a latent uncertainty state drives both, prompting the use of generative models. I discount these causal mechanisms. First, VIX is a forward-looking market-based measure. Investors trade options, the prices of which signal their expectations of the volatility of the underlying security. VIX itself cannot be subject to exogenous shocks, because only a change in trading behavior of market participants broadly can shift VIX. Any change in VIX is likely propogated through changes in investors' information set. Second, VIX itself rarely receives news coverage. Investors or policymakers may change their behavior in response to fluctuations in VIX, but those actions should be reflected in news coverage so this channel is non-unique. Consequently, I don't pursue the technique of inverting fitted generative models[21] to obtain estimates of VIX as suggested in Gentzkow et al (2019).

[20]This paper is an exercise in pure prediction. Causal questions, such as how specific words drive changes in the expectations of forward-looking investors and thus implied volatility, are not considered. Model interpretability is important but the primary criteria for model selection is out-of-sample predictive performance. See Appendices 11.3 and 11.4 for details on error measures.

[21]An approach is to estimate a topic model over the entire WSJ corpus, regress VXO post-1986 on

5.2 Regression Models

An entire suite of high-dimensional regression methods is applicable to text. These methods predict a continuous dependent variable from a set of covariates and are suited to the case of more covariates than observations[22]. I evaluate the performance of penalized linear models, dimensionality reduction models, support vector machines, tree-based methods, and deep learning models. Estimated implied volatility at t is a function of month t's WSJ text and a set of parameters, $\hat{\theta}$.

$$\begin{aligned} \widehat{\mathrm{IV}}_t &= f(\vec{\theta}, c_t) \\ E(\mathrm{IV}_t | c_t) &= f(\vec{\theta}, c_t) \end{aligned} \tag{7}$$

A multi-layer perceptron is a fully connected feed-forward neural network with multiple hidden layers. A support vector regression extends the concept of a margin in support vector machines to select coefficients by minimizing a loss function in which only residuals with an absolute value in excess of a specified constant matter. The parameter C is critical to mitigate overfitting. The stochastic gradient descent regressor uses stochastic gradient descent to estimate various regularized linear models. I use two implementations of gradient boosting regression, one from sci-kit learn and another from LightGBM. Both operate by forming an ensemble of decision trees. Principal components regression and partial least squares construct new covariates by projecting data onto a lower-dimensional subspace. I provide a brief treatment of each model, covering the algorithm and key hyperparamters. All exposition is based on Hastie et al (2017) unless otherwise noted.

the monthly panel of topic distributions, obtain coefficients, and predict VXO pre-1986 using fitted coefficients and the pre-1986 panel. Lack of topic persistence and collinearity make estimation difficult.

[22]Ordinary Least Squares is not feasible because the number of covariates vastly exceeds the number of samples. The covariate matrix is rank deficient, so the Gram matrix is singular, and the OLS estimator is infeasible. Note, even if the feature space was reduced to the sample size, i.e. the case of $p \approx n$ instead of $p >> n$, there are numerical stability issues associated with $(X'X)^{-1}$.

Support Vector Regression: Minimize the function

$$H(\beta, \beta_0) = \sum_{i=1}^{N} V(y_i - \beta_0 - x_i'\beta) + \frac{\lambda}{2}||\beta||^2 \tag{8}$$

for ϵ-insensitive error

$$V_\epsilon(r) = (|r| - \epsilon)\mathbf{1}_{(|\mathbf{r}|\geq\epsilon)} \tag{9}$$

which ignores errors less than ϵ. In some formulations, $c = \frac{\lambda}{2}$. The vector of coefficients β is a function of positive values $\hat{\alpha}_i^*$,$\hat{\alpha}_i$ that solve a quadratic programming problem

$$\hat{\beta} = \sum_{i=1}^{N}(\hat{\alpha}_i^* - \hat{\alpha}_i)x_i \tag{10}$$

and only a subset of which (support vectors are the associated data) are nonzero. Key hyperparameters are regularization parameter c and ϵ which controls the insensitive zone.

Gradient Boosting Regression: An iterative procedure as follows:

1. Initialize $f_0(x) = \arg\min_\gamma \sum_i L(y_i,\gamma)$

for loss function L. This is an initial constant value prediction. Loss can be squared loss for regression, $(y_i - \gamma)^2$, thus $\gamma = \bar{y}$.

2. For iterations $m = 1$ to M:

a. For $i = 1...N$ observations compute the negative gradient to obtain pseudo-residuals:

$$r_{im} = -\left[\frac{\partial L(y_i, f(x_i)))}{\partial f(x_i)}\right]_{f=f_{m-1}} \tag{11}$$

Take the derivative of the loss function and evaluate it at previous prediction f_{m-1} and multiply by negative 1.

b. Fit a regression tree to r_{im} to obtain terminal regions R_{jm} for $j = 1,2,...,J_m$.

There are J_m leaves in tree index m.

c. For $j = 1,2,...,J_m$ compute:

$$\gamma_{jm} = \arg\min_{\gamma} \sum_{x_i \in R_{jm}} L(y_i, f_{m-1}(x_i) + \gamma) \tag{12}$$

where γ_{jm} is the value that minimizes the loss function at each terminal mode, or effectively the predicted value of the regression tree.

d. Update $f_m(x) = f_{m-1}(x) + \sum_j \gamma_{jm} \mathbf{1}_{(\mathbf{x} \in \mathbf{R_{jm}})}$, meaning increment the prior prediction if a given x is in the terminal region R_{jm}.

3. Set $\hat{f}(x) = f_M(x)$.

The key hyperparameters are M, which is the number of iterations and trees created, and J_m, the size of each of the constituent trees.

Multi-Layer Perceptron Regression: A multi-layer feed-forward neural network that predicts a continuous value. The loss function is squared error, and the architecture consists of simple fully-connected hidden layers. Key hyperparameter choices are the hidden layer sizes, passed as a tuple (e.g. (100,50,10) represents 3 hidden layers with 100, 50, and 10 nodes respectively), activation function, learning rate, backpropagation method (SGD by default), and α, the penalty magnitude for the L2 regularization term.

Stochastic Gradient Descent Regression: A penalized linear model implemented with stochastic gradient descent:

$$\hat{\beta} = \arg\min_{\beta} L(\beta_0, \beta) + n\lambda \sum_{j=1}^{p} \kappa_j(|\beta_j|) \tag{13}$$

for n observations, p covariates, penalty magnitude $\lambda > 0$, and cost functions κ_j that penalize deviations of the parameters from zero such as the L1 or L2 norms (Gentzkow et al 2019). By default, the loss is squared error and the penalty is the L2 norm. Stochastic gradient descent estimates the gradient of the loss function at each sample at a time and the parameters are updated per a decreasing strength schedule. The penalty magnitude

λ is represented as α in the sci-kit learn documentation. Key hyperparameters are the choice of loss function, penalty (L2, L1, or combination), and penalty magnitude.

Principal Components Regression: Regress $\mathbf{y}$ on derived orthogonal principal components $\mathbf{z}_1....\mathbf{z}_M$ for $M \leq p$ original standardized covariates. Each $\mathbf{z}_m$ is a linear combination of the covariates, $\mathbf{z}_m = \mathbf{X}v_m$.

$$\hat{\mathbf{y}}^{pcr}_{(M)} = \bar{y}\mathbf{1} + \sum_{m=1}^{M} \hat{\theta}_m \mathbf{z}_m \tag{14}$$

for $\hat{\theta}_m =< \mathbf{z}_m, \mathbf{y} > / < \mathbf{z}_m, \mathbf{z}_m >$. The key hyperparameter is M principal components[23].

Partial Least Squares: Standardize each covariate $\mathbf{x}_j$ to have mean 0 and variance 1. Compute $\hat{\psi}_{1j} =< \mathbf{x}_j, \mathbf{y} >$ for each j and derived input $\mathbf{z}_1 = \sum_j \hat{\psi}_{1j}\mathbf{x}_j$. Regress $\mathbf{y}$ on $\mathbf{z}_1$ to obtain $\hat{\theta}_1$ and orthogonalize $\mathbf{x}_1...\mathbf{x}_p$ with respect to $\mathbf{z}_1$. Repeat to derive all directions. The key hyperparameter is M, the number of PLS directions .

All hyperparameters are tuned using 5-fold cross-validation. This procedure entails partitioning the training data into five folds, fitting the model on four folds and testing it on the fifth, varying the heldback fold to get error measures for each fold, and then averaging over test errors to get a measure of test error per hyperparameter configuration.

5.3 Classification Models

Text classification is the task of predicting categorical labels, either binary or multiclass, from text data. A regression problem can be transformed into a classification problem by discretizing the dependent variable into categorical buckets[24]. I bin observed VXO values into terciles, representing periods of high, intermediate, and low volatility, and train several classifiers to predict volatility period type instead of level. This serves as a

[23]Note, <x,y> is the inner product between vectors x and y, also noted as $x^T y$ or $\sum_{i=1}^{n} x_i * y_i$.

[24]For example, a model tasked with predicting a value in the XY plane can be repurposed to predict which quadrant the value will fall in.

reference point or upper bound for the plausible performance of any regression model and a litmus test for the plausibility of estimated implied volatility. Predicting the exact value of implied volatility is a harder task than predicting one of the terciles. For classification, I use K-nearest neighbors, support vector machine, decision tree, random forest, multi-layer perceptron, AdaBoost, Gaussian naive Bayes, and a stochastic gradient descent classifier.

5.4 Training and Testing Methodologies

Splitting a dataset into train, validation, and test subsets is a common method to assess a model's performance in a supervised learning task. The two primary methods to split the dataset are a randomized split and a temporal or otherwise deterministic split. A randomized split is often inadvisable for time series data that exhibits autocorrelation because it risks data leakage[25]. Data leakage is the illegitimate introduction of information about the target of a data mining problem into the data mining process. Models beset by data leakage generalize poorly and overestimate expected performance (Kaufman et al 2011). In a times series context, a model to predict a future value at $t + 1$ will not have access to future values $t + 2$,....,$t + \tau$. A randomized split allows the model to learn to use values from $t + \tau$ to predict prior values at $t + 1$. Implied volatility exhibits autocorrelation; periods of high VXO are often followed by high VXO, and low VXO by low VXO. Thus the risk is especially pertinent because inference based on text features becomes indistinguishable from inference based upon trends in the times series[26]. High out-of-sample R^2 could be evidence of the model acting as an interpolation and predicting a value of implied volatility based upon the most textually proximate training sample. I

[25]An upside is that when used with random shuffling, it destroys the time series structure of data. This can force certain models to learn the relation between particular text features and implied volatility instead of just trends in implied volatility

[26]Imagine implied volatility in the training set is high at $t + 1$ and $t + 3$, and the model has the straightforward task of predicting implied volatility at $t + 2$.

conduct robustness checks for the top performing model that utilizes a random data split by non-parametric bootstrapping of test errors with repeated random shuffling[27]

A temporal split is also undesirable given that VXO is generally much higher after 1996 than before[28]. Regardless of the training period, any model will necessarily overfit to either high or low implied volatility. If the training period is fundamentally a higher VXO period, and the model overfits to the training sample, one would expected predicted implied volatility from July 1889 to December 1985 to be higher as a result. A logical solution is to set the test set as chronologically after the training set, ensure the training set is considerably larger than the test set, and predict forward, however this paper is an exercise in predicting over the pre-1986 period. Data-leakage however is a broader concern. Any methodology that trains on post-1986 data to predict over pre-1986 values uses information that people in the prior period couldn't access. Investors during the period July 1889 to December 1985 couldn't condition their expectations of uncertainty on an information set from the period January 1986 to December 2009. A model trained on the 1986.01+ period to predict over a chronologically prior period assumes future information can be used to estimate past implied volatility, a concern Manela and Moreira dismiss owing to the text mining approach of counting n-gram frequencies[29].

Dwarfing all these considerations is the quandry of insufficient data. There are at most 287 training samples from January 1986 to December 2009 (Jan. 1992 is omitted

[27]See Appendix 11.16 for this exercise.

[28]VXO is on average 23.24 in the 1996.01-2009.12, with a median of 23.01, and on average 18.75 with a 75th percentile of 21.35 during the 1986.01-1995.12 periods. Half of the observed VXO values in the 1996.01-2009.12 period are greater than at least 75 percent of the observed VXO values in the 1986.01-1995.12 periods

[29]They argue that journalists would have to change the manner in which particular shocks are covered because they have knowledge of the prior impact of shocks, such as the WSJ in 2008-2009 covering stock market movements more during the Great Financial Crises on account of the role the Crash of 1929 played in the Great Depression (Manela and Moreira 2017). They state this is not the case. Summary statistics on the count of unique n-grams and a review of historical front-pages show the nature of news coverage varies, so the concern is not so easily dismissed.

from the replication kit) to train a model on. Using only half for training risks strongly overfitting to the implied volatility levels and macroeconomically relevant events unique to the training period[30]. A model cannot learn relationships between coverage of key events and implied volatility if it is not trained on text from such events; the issue is noteworthy when using 80 percent of training data and unavoidable when using half.

Table 7: Train/Test Configurations

Config.	Train Set	Test Set	Method	Dependent Variable
1	1986.01-2009.12	1986.01-2009.12	80/20 Random Split	VXO
2	1927.07-2009.12	1927.07-2009.12	80/20 Random Split	Realized Vol.
3	1996.01-2009.12	1986.01-1995.12	Backwards	VXO
4	1986.01-1995.12	1996.01-2009.12	Forwards	VXO
5	1996.01-2009.12	1986.01-1995.12	Backwards	Realized Vol.
6	1986.01-2009.12	1986.01-2009.12	80/20 Random Split	VXO Tercile
7	1990.10-2009.12	1986.01-1990.09	Backwards	VXO
8	1986.01-2005.03	2005.04-2009.12	Forwards	VXO

I test each text regression model for 1-5. 6 is the classification setup. I test 7 and 8 on only gradient boosting regression. Manela and Moreira use 3 for VXO and 5 for validatio on realized volatility.

I use multiple train/test configurations. The first uses an 80/20 random split with VXO as the dependent variable and the second uses a random split with S&P500 realized volatility as the dependent variable and the maximum time series of July 1927 to December 2009. For all methods using a random split, each model is tested on the same partition of the data to permit performance comparisons. Configuration 3 is used by Manela and Moreira and involves training over January 1996 to December 2009 and testing backwards over January 1986 to December 1985. Configuration 4 involves training over January 1986 to December 1985 and testing forwards over January 1986 to Decem-

[30] I explore rolling training strategies as well. These strategies involve training on window of n years, and predicting over year $n+1$, and rolling forward or backward through the 1986.01-2009.12 period. Models trained in this manner exhibit severe overfitting. See Appendix 11.11 for discussion.

ber 2009. Configuration 5 is Manela and Moreira's validation procedure, in which they train and test a model on realized volatility as done in configuration 3, but compare the predictions over the pre-1986 period to historical realized volatility. Configuration 6 is the classification exercise to predict VXO terciles. Configurations 7 and 8 use a temporal 80/20 forward and 80/20 backward set up, which trains on 80% of the data and tests on the temporally first or last 20%. This separates the effect of the particular random split from the effect of insufficient training data. All configurations except 2 and 5 use VXO as a dependent variable. I test configurations 1-6 on all text regression models and limit configurations 7 and 8 to discussion of gradient boosting regression.

Each of the aforementioned concerns is meritorious. I subsequently introduce the concept of transfer learning and argue that owing to these concerns, and other reasons, only transfer learning is a viable methodology[31].

[31]The optimal train-test configuration also bears on extensions of this work that incorporate the 13 years of text and implied volatility data after the end of the sample, 2009.12, used in this paper.

5.5 Variance Risk Premium

Historical plausibility of an estimated implied volatility series involves more than an eye-test of large deviations corresponding to major events. Many macroeconomic variables can be used as an assessment by establishing their relationship with observed implied volatility post-1986 and comparing to the relationship with pre-1986 estimates. An empirical test of the quality of estimated historical implied volatility is the variance risk premium. An asset's time-varying variance risk premium is the difference in ex-ante expectation of return variance under the risk-neutral Q probability measure and the statistical expectation of return variance under the physical P probability measure (Zhou 2018). It is not directly observable, but is estimated from the difference between model-free implied variance and lagged model-free realized variance.

$$\begin{aligned}
\mathrm{VRP}_t &\equiv E_t^Q(\mathrm{Var}_{t,t+1}) - E_t^P(\mathrm{Var}_{t,t+1}) \\
\widehat{\mathrm{VRP}}_t &\equiv \mathrm{IV}_{t,t+1} - \mathrm{RV}_{t,t+1} \\
IV_{t,t+1} &= \frac{\mathrm{VIX}_{t,t+1}^2}{12} \\
RV_{t,t+1} &= \sum_{j=1}^{n} \left(\ln(\frac{p_{t+j/n}}{p_{t+(j-1)/n}}) \right)^2
\end{aligned} \tag{15}$$

Option implied variance is monthly VIX squared and divided by 12. Realized variance, measured over a discrete interval t to $t+1$, is the sum of squared log returns. Many papers document the predictive power of the variance risk premium in forecasting equity returns (Bollerslev et al 2009). VIX is a weighted average of the prices of call and put options on the S&P500, wherein out of the money put options receive more weight. On average, the variance risk premium is positive. This indicates risk-averse investors pay a premium to hedge against time-varying volatility. A curious stylized fact about the variance risk premium is that it is on occasion negative in extreme crises like the outbreak of Covid in March 2021 and the onset of the Great Financial Crisis in September 2008. A decreasing

Figure 2: Variance Risk Premium, 1990.01-2021.12

variance risk premium suggests investors become oddly less averse to risk during crises, but the stylized fact remains[32]. From January 1990 to December 2021, the S%P500 variance risk premium averaged 15.09 with a standard deviation of 29.66, a minimum of -403.40 and maximum of 115.85, and 25th, 50th, and 75th percentiles of 7.47, 12.85, and 22.73, respectively. Option implied variance and realized variance have a correlation of 0.78 during the period. With the aforementioned stylized facts and summary statistics from the measured variance risk premium in mind, I compute the variance risk premium for leading candidates of historical implied volatility. I use formal non-parametric tests guided by the intuition that basic facts, such as the on average positive risk premium or rough correlation between option implied and realized variance, should hold up in the July 1889 to December 1985 period using estimated implied volatility.

[32]See Appendix 11.12 for more details on crises in which the premium remained positive.

6 Conventional Training Results

For each training and testing configuration, I evaluate the gradient boosting, multi-layer perceptron, penalized linear, support vector, principal components, and partial least squares regressors on each variant of the document-term matrices. I report the best performing model and document-term matrix combination per train/test configuration.

Table 8: Regression and Classification Model Performance

Reg.	Model	Vectorizer	Features	Test Sample	RMSE	R^2
1	Gradient Boosting	Count	342,155	58	4.96	0.48
2	Multi-Layer Perceptron	Count	5,000	198	6.36	0.65
3	Penalized Linear	TFIDF	342,155	119	7.45	-0.11
3	Support Vector*	Count	460,891	119	7.45	-0.11
4	Support Vector	TFIDF	342,155	168	9.62	-0.11
5 Train	Gradient Boosting	TFIDF	5,000	168	3.40	0.89
5 Test	Gradient Boosting	TFIDF	5,000	119	9.34	-0.49
5 Predict	Gradient Boosting	TFIDF	5,000	702	10.04	-0.15
Class.	**Model**	**Vectorizer**	**Features**	**Test Sample**	**Acc.**	**F1**
6	RBF Support Vector	Count	5,000	58	0.71	0.71

Note: Out of sample unless otherwise noted. * is Manela and Moreira's SVR. Configuration: 1: 1986.01-2009.12 80/20 Random Split on VXO, 2: 1927.07-2009.12 80/20 Random Split on SP500 Realized Vol, 3: Train 1996.01-2009.12 and Test 1986.01-1995.12 on VXO, 4: Train 1986.01-1995.12 and Test 1996.01-2009.12 on VXO, 5: Train 1996.01-2009.12 and Test 1986.01-1995.12 on SP500 Realized Vol.,6: 1986.01-2009.12 80/20 Random Split on VXO Tercile.

The document-term matrices differ in their vectorizer and number of text features, ranging from the original vocabulary of 460,981 n-grams to the smallest vocabulary of 5,000 unigrams. After identifying the best performing model and document-term matrix per training method[33], I perform 5-fold cross-validation on each top performing model to ex-

[33]A comprehensive approach involves cross-validation for every model, document-term matrix, and training method combination, but I found the improvements per model marginal. Default parameter choices from sci-kit learn tend to be well-suited for this task.

tract incremental gains in performance[34]. For brevity, I report out-of-sample root mean squared error and R^2.

Amongst the conventional training methods, the 80/20 randomized split approach is most conducive to high test performance and a gradient boosting regressor achieves out-of-sample R^2 of 0.476. A multi-layer perceptron regressor, using S&P500 realized volatility as the dependent variable and an 80/20 split on the 1927.07-2009.12 sample, achieves the highest R^2 of 0.6461. Configuration 3 is the original train period of 1996.01-2009.12 and test period of 1986.01-1995.12 used in Manela and Moreira. A penalized linear model achieves marginally better R^2 of -0.1096 and RMSE of 7.448 compared to the author's SVR which achieves R^2 of -0.111[35] and RMSE 7.454 The high test error and negative R^2 indicate extremely poor fits. High R^2 means the model correctly predicts movements in VXO from test data, and low RMSE indicates the model predicts close to the true VXO value. The same support vector regression performs the best when the test period is chronologically after the train period. Training configuration 5 is a validation exercise for the long out-of-sample prediction period 1889.07-1985.12, in which a model is trained on 1996.01+ data and tested on 1986.01-1995.12. Its predictions over 1889.07-1985.12 are compared to available data on realized volatility during that period (Manela and Moreira 2017). Compared to Manela and Moreira's support vector regression with an RMSE of 16.456 and 19.616 for the test and predict samples, respectively, the gradient boosting regression achieves considerably lower RMSE for the test and predict samples of 9.336 and 10.042, respectively. Gradient boosting performs reasonably well over the predict period and is best suited, out of all off the shelf models, to generate an estimate of

[34]Optimal hyperparameters are GBR(α=0.05, max depth 1, 500 estimators, 30 leaves), Penalized Linear(α=0.01, Intercept, L2), MLPR(3 hidden layers: 512,64,32, ReLu activation, Adam optimizer), SVR(C $\approx$50.74, $\epsilon \approx 3.49$), PCR(standard scaling, N=15), PLS(standard scaling, N=5).

[35]Despite the true negative R^2, this result was published in the JFE. Their model does worse than predicting average VXO. Penalized linear performance is subpar but still an improvement over the best existing approach, especially considering the data reduction.

implied volatility. I achieve all results with considerably fewer than the 460,891 n-grams in the original vocabulary. Only 5,000 n-grams, or just over one percent of the original vocabulary, is needed to achieve superior performance. This means many text features in the WSJ corpus characterize events that do not bear on implied volatility at all. Lastly, I report classification results for VXO terciles. A support vector machine achieves test accuracy of 0.708 and F1 of 0.709. By construction, the number of classes for high, medium, and low VXO is equal. This provides an intuitive upper bound on performance.

Earlier I noted that a randomized split of the training data may result in spuriously high out-of-sample performance. To investigate the effect of various training methods, I use a gradient boosting regressor and count-vectorized document-term matrix with 342,155 features. Forward training methods 3 and 7 are alike except 3 has 62 fewer training samples than 7. Backward methods 4 and 8 are alike except 4 has 111 fewer training samples than 8. Increased training reduces model RMSE by 1.01 for backwards methods and 0.31 for forwards methods. R^2 increases by 0.78 for backwards and 0.86 for forwards. Given similar train and test sample sizes, a randomized split unquestionably can inflate test R^2. Bootstrapping shows the measure is stable, so I proceed with its use.

Table 9: Gradient Boosting Regression

Config.	Description	Train/Test Sample	RMSE	R^2
1	Randomized Split	229/58	4.96	0.48
3	Backwards	168/119	11.26	-1.53
4	Forwards	119/168	12.24	-0.79
7	Backwards	230/57	10.25	-0.35
8	Forwards	230/57	11.93	0.07

Note: Out of sample unless otherwise noted. Configuration: 1: 1986.01-2009.12 80/20 Random Split on VXO, 3: Train 1996.01-2009.12 and Test 1986.01-1995.12 on VXO, 4: Train 1986.01-1995.12 and Test 1996.01-2009.12 on VXO, 7: Train 1990.10-2009.12 and Test 1986.01-1990.09 on VXO, 8: Train 1986.01-2005.03 and Test 2005.04-2009.12 on VXO.

6.1 Evidence from the Estimated Variance Risk Premium

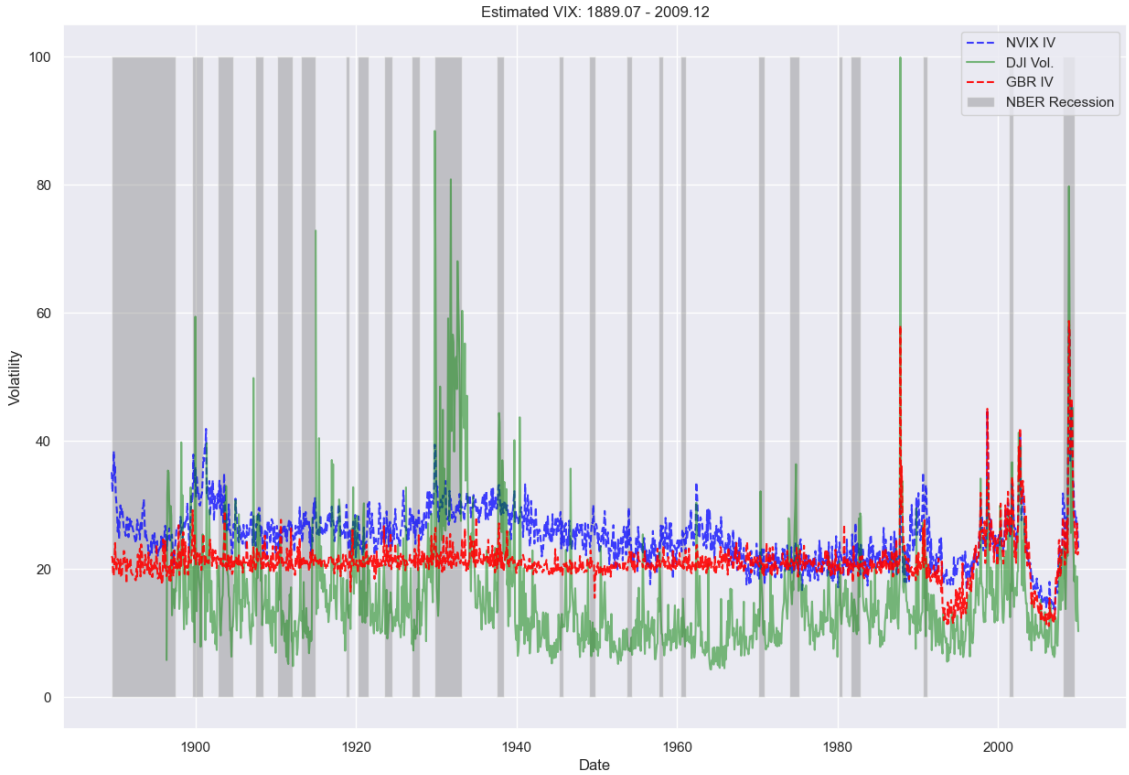


Figure 3: Estimated Implied Volatility and Realized Volatility, 1889.07-2009.12

When plotted against Dow Jones Index realized volatility, implied volatility estimates from the gradient boosting regression model and Manela and Moreira's NVIX appear implausible. Both estimates co-move with realized volatility during the post-1986 training period but gradient boosting implied volatility is effectively flat, with no response to events like the Crash of 1929 and muted responses to financial panics early in the 20th century. NVIX misses Black Monday in 1987 and behaves curiously at the turn of the 19th century. The models fit to text features and events in the 1986.01-2009.12 period that are either not prevalent or unlike those pre-1986. Despite the validation exercise on realized volatility and the lower test errors for the gradient boosting model, its historical implied volatility

estimates do not align with the historical record of major shocks during the 20 century. I compute the estimated variance risk premium for both and compare summary statistics.

Table 10: NVIX and Gradient Boosting Estimated Variance Risk Premium

	Gradient Boosting 1896.05-2009.12	NVIX (MM 2017) 1896.05-2009.12	Measured VRP 1990.01-2021.12
Count	1359	1359	384
Mean	10.67	25.96	15.10
Std.	47.48	47.99	29.66
Min.	-596.42	-753.87	-403.40
p25	10.56	21.29	7.47
p50	21.56	33.09	12.85
p75	27.87	45.41	22.73
Max.	111.94	116.15	115.85
$\rho_{IV,RV}$	0.47	0.43	0.78

Note: $\widehat{\text{VRP}}_t = \text{IV}_t - \text{RV}_t$ for $IV_t = \text{VIX}^2/12$ and realized variance equals the sum of squared log returns. NVIX is an estimate of implied volatility from Manela and Moreira (JFE 2017).

Summary statistics are not prima facie evidence against either estimate of implied volatility. However, the considerably lower correlation between option implied variance and realized variance - 0.47 for gradient boosting and 0.43 for NVIX- compared to 0.78 from 1990.01-2021.12, undermines their credibility. The gradient boosting average at 10.56 is closer to the measured average of 15.10. I perform Kolmogorov-Smirnov tests and reject the null hypothesis that both gradient boosting and NVIX variance risk premium series are drawn from the same distribution as the measured variance risk premium.

Table 11: Kolmogorov-Smirnov Tests

Sample	Null	Alternative	Test Statistic	P-Value
Gradient Boosting	F_{GBR},$F_M \in F$	F_{GBR},$F_M \notin F$	0.269	0.00
NVIX	F_{NVIX},$F_M \in F$	F_{NVIX},$F_M \notin F$	0.490	0.00

7 Transfer Learning

Conventional approaches to estimating historical implied volatility that involve calibrating a model to the post-1986 period for which options price data is available and then predicting over the long pre-1986 sample are invariably deficient. The predict sample is too large relative to the maximum training sample, the training sample has qualitatively and quantitatively insufficient data, and each type of partition of the training sample has deep flaws. Back predicting itself is a dubious endeavor because it imputes information not available in bygone eras to investors expectations of uncertainty. Foregoing decades of data on realized volatility sacrifices crucial history in order to preserve the meaningful distinction between implied and realized volatility.

Transfer learning solves each of the aforementioned problems[36]. A facile account of historical implied volatility is that it resembles realized volatility but is smoother, on account of being a forward-looking measure, and generally higher than realized volatility, on account of investors paying a premium when purchasing options to hedge against stochastic volatility. Although imprecise, this account captures the intuition of using transfer learning to take a model trained to predict realized volatility and redeploy it to predict historical implied volatility. I outline the theoretical rationale for transfer learning and procedural aspects.

7.1 Motivation

Qualitatively and quantitatively, the historical records of major financial crises, pandemics and contagions, political scandals, wars and geopolitical crises, assassinations, terrorist attacks, and natural disasters (Bloom 2009) during the periods of July 1889

[36] An additional benefit pertains to the stability of the meaning of text feature. The bigram "Japanese Navy" in 1942 had a much more negative connotation than it did in 1992, so a model exposed to text from the decades prior to 1986 can learn such distinctions.

to December 1985 and January 1986 to December 2009 differs considerably. News coverage during the post-1986 period cannot reflect events that didn't happen and cannot meaningfully resemble events of unquestionably greater magnitude pre-1986[37].

Amongst pandemics and contagions, there is no post-1986 analogue to the Spanish Flu of 1918, in which at least an estimated 50 million people worldwide and 675,000 in the US died. The outbreak of H2N2 in 1957 killed 1.1 million people worldwide and 116,000 in the US, and an outbreak of H3N2 in 1968 killed 1 million people worldwide and 100,000 in the US. In contrast, the 2009 H1N1 pandemic killed an estimated 12,469 in the US. In 1900, the Galveston hurricane killed an estimated 6,000 people; in 2005, Hurricane Katrina killed over 1,200 despite considerably higher costs. San Francisco in 1906 was the site of the deadliest earthquake US history. World War I and World War II have no historical analogues in terms of number of belligerents and theaters of conflicts, financial costs, death toll, and levels of national mobilization. Military conflicts post-1986, such as the Gulf War or Iraq War, involved no more than two principal belligerents and did not induce persistent economic disruption. Attacks on US military installations during the War on Terror were simply not reported in the same manner as Pearl Harbor on December 7th, 1941. Two American presidents – William McKinley and John F. Kennedy - were assassinated between 1889 and 1986, and none since. Global thermonuclear war was never more proximate than during the Cuban Missile crises of November 1962; as the US enjoyed its post-Cold War peace dividend, the specter of nuclear conflict largely disappeared from news coverage. Coverage of nuclear proliferation post-9/11 was qualitatively different as well. In contrast, terrorist attacks became more pervasive and deadly in recent decades, with the attacks of Sept. 11th, 2001, inducing huge volatility. Financial crises were more frequent post-1986 but their origins differed. The Panic of 1893 was precipitated by a

[37]See Appendix 11.15 for a table of major historical events in both periods.

drop in gold reserves, a metric largely disconnected from financial markets since 1971. Efforts to corner stock of Northern Pacific Railway caused the Panic of 1901, and the first worldwide financial crisis of the 20th century, the Panic of 1907, began with a failed attempt to corner the stock of United Copper. Bank runs were the primary transmission mechanism. The Crash of 1929 is often compared to the Great Financial Crisis, but the latter involved the housing market and complex derivatives and the former speculation and purchases of stock on margin. During the 1990s, crises were associated with foreign devaluations, such as the 1994 Peso Crisis, or defaults, such as the 1998 Russian default and corresponding implosion of LTCM. Black Monday in 1987 and the Great Financial Crisis of 2008-2009 highlight a pattern of more frequent and severe financial shocks, and of varied origin, with greater adverse macroeconomic consequences. A model exposed to only financial shocks pre-1986 would find post-1986 shocks foreign, and vice-versa.

Bifurcating news coverage of key events to train and test a model necessarily compromises the information set a model is exposed to. Only an approach that accesses both the pre-1986 and post-1986 information sets can account for disparities in the quantity and qualitative aspects of major disasters and shocks to uncertainty.

7.2 Procedure

Transfer learning is a machine learning technique that consists of taking a model pre-trained for tasks in one domain and deploying it, in a calibrated manner, to another similar domain (Aggarwal 2018). The underlying premise of transfer learning is that feature representations learned in early hidden layers can be applied to other datasets in similarly situated domains. Each hidden layer in a neural network outputs a transformed feature representation of the data. Consequently, the outputs of earlier layers can be thought of as primitive characteristics of the data, with the sophistication of representation increas-

ing in the number of hidden layers as a result of non-linear activation layers[38] (Aggarwal 2018). In this application, early layers of a neural network trained to predict realized volatility learn general levels of volatility from text data, such as whether volatility is low, moderate, or elevated, and subsequent layers learn the specific relationships between text features like war and stock and precise levels of realized volatility. The classification exercise is informative. A neural network with a single hidden layer can accurately classify terciles of volatility, but multiple hidden layers are necessary to predict specific levels of volatility.

In practice, transfer learning can be as simple as replacing the sigmoid activation function in the single neuron of the final layer of a pretrained neural network binary classifier with a linear activation function to create a neural network regression model. More commonly, transfer learning involves downloading a large pretrained state-of-the-art general model, such as ImageNet, freezing a subset of the layers and training the remaining layers on additional, domain-specific data. Freezing the weights of the early layers preserves the primitive but generalizable features learned in the pre-trained model, such as edges. The later hidden layers are initialized with random weights and trained on the new domain-specific data, learning the complex features that differentiate the new samples from those the pretrained model was exposed to. The first two hidden layers of a neural network learn to map text features to rough realized volatility levels, and the later layers are retrained to learn the nuances that differentiate implied volatility from realized volatility.

The assumption of primitive but generalizable features is paramount. In computer vision, the notion that features like edges are salient across a range of datasets is eminently

[38]For example, an image classifier trained to recognize dogs will learn edges outlining the general shape of a four-legged animal in early hidden layers, whereas the later hidden layers will learn more complex geometric objects associated with specific features like the nose or teeth.

plausible. I hypothesize that exogenous shocks to investors' information sets, represented with text features from historical news coverage, bears on trading activity which manifests in changes to realized volatility. The shocks also bear on forward-looking expectations of uncertainty manifesting in changes to implied volatility. Evidence of this relationship is the contemporaneous correlation between realized and implied volatility. An example of the mechanism is as follows. An exogenous shock occurs, such as a terrorist attack. Investors ingest news text, learn of the shock, and immediately trade to minimize their exposure to positions adversely affected by the shock. Realized volatility increases. Investors update their expectations assuming persistence of the effects of the shock or an impending correlated shock, such as a military response, placing a premium on hedging options and driving up implied volatility. However, they have already exited susceptible positions so realized volatility falls.

Data sparsity informs the number of hidden layers for which pretrained model weights are preserved, as does the structure of the pretrained neural network; with more domain-specific data available, more layers can be trained. Data sparsity is also a principal motivation for transfer learning, as it requires considerably less training data than training a model from scratch to achieve comparable performance. VIX is available from 1986 onward, but measures of realized volatility date back to 1896. The larger range for realized volatility as a dependent variable permits use of decades more worth of text in a supervised learning model, making backdating VIX a prime candidate for transfer learning. Learning across tasks, characterized by Thrun and Pratt (2012) as learning to learn, involves always reusing trained models to improve performance on new tasks. I further hypothesize that a neural network regressor for volatility will improve its performance on both implied and realized volatility because it is applied to predicting both, owing to increased exposure to major historical events as outlined above.

I utilize the following procedure for transfer learning:

1. Train a neural network regressor to predict realized volatility using the 1896.05-1985.12 WSJ corpus and DJIA realized volatility as a dependent variable, splitting 90% of data for training and 10% for validation.

2. Freeze the first two hidden layers of the neural network so that their learned weights cannot be updated during backpropogation, and reinitialize the remaining layers with random weights.

3. Retrain the neural network to predict implied volatility using the 1986.01-2009.12 WSJ corpus and VXO implied volatility as a dependent variable, using either 80% of the data for training and 20% for testing or a temporal train/test split.

4. Unfreeze all layers of the model and retrain the neural network again as in step 3 but with a minute learning rate to preserve representations and weights learned in step 3.

5.. Predict implied volatility over the entire 1889.07-2009.12 WSJ corpus.

Step 4 is known as fine-tuning. I conduct several robustness checks, detailed below.

7.3 Neural Architecture

I develop a simple feed-forward neural network regressor to predict realized volatility and then use transfer learning to predict implied volatility from the same basic architecture. The network consists of an input layer, several hidden layers, and an output layer. Several hidden layers broaden the capacity of the neural network; the layer sizes are 128, 32, and 8, permitting a large model of non-linear relationships between text features and volatility measures. I utilize a rectified linear unit activation function for all but the output layer, which has a linear activation. More hidden layers and nodes result in more parameters to be learned[39]. An excess of free parameters causes the model to "memorize"

[39]There are 640,128 (5,000 features * 128 neurons) parameters associated with the first hidden layer. This neural network is quite modest, but I found that it outperformed more complicated architectures.

nuances which prevents generalizing to unseen test data, a phenomena known as overfitting. Evidence of overfitting is near perfect train error and high test error. Overfitting in this application means the model simply predicts for a test sample the level of volatility associated with the most textually similar training sample instead of learning to relate events and uncertainty shocks to realized or implied volatility through text features.

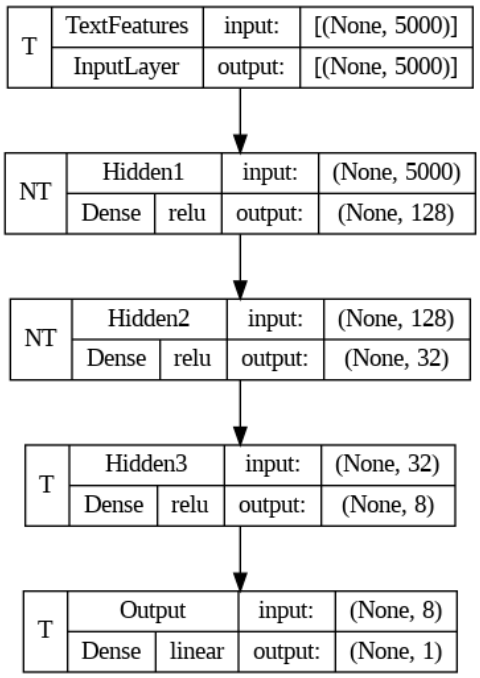


Figure 4: Regressor Architecture

To prevent overfitting, I implement early stopping and use exponential learning rate decay. I found that dropout layers[40] and batch normalization[41] decrease performance. Early stopping is a form of regularization in which a portion of the training data is heldback for validation, and when the validation set error rises despite falling on the training set,

[40]Dropout is a regularization method that mimics the effect of ensemble training which uses combinations of trained neural networks. A dropout rate of 0.20 means for each fully-connected layer that nodes drop out with probability 0.20 to compensate for the larger network size

[41]Batch normalization is a technique to prevent vanishing or exploding gradients, and consists of adding a normalization layer between hidden layers to create features with similar variance. The output of a BN layer has the same mean and standard deviation over each mini-batch of training instances. Normalization can be done before or after applying an activation function

the gradient descent approach ends. As a result, the parameter space shrinks to be closer to the initial parameter values. Early stopping requires no further hyperparameters and the upside of reduced overfitting often outweighs the loss of training data, hence it is a common procedure. Early stopping prevents the model from memorizing textual features of events of the training period and looking for those exact features in the test period to predict volatility. I use validation accuracy and a patience of 5 epochs as criteria for early stopping. A low learning rate results in slow convergence, whereas a high initial learning rate can result in divergence or oscillation around the optima. A decaying learning rate results in a high initial learning rate and lesser later learning rate, capturing the benefits of both. I utilize an initial $\alpha = 0.01$, 10,000 steps, and a rate of 0.9. For fine-tuning, the initial learning rate is $\alpha - 0.001$. A neural regressor's output layer has a single node with a linear activation to output the predicted continuous value.

8 Transfer Learning Results

I perform steps one through four from the aforementioned procedure for each of the three implied volatility training configurations, using the 5,000 text feature TFIDF vectorized document-term matrix. In step 1, the neural network achieves peak out-of-sample R^2 of 0.77 and RMSE of 1.05, indicating a good fit to the DJIA realized volatility data. The 10% test sample is smaller to expose the model to more events during the 1896.05-1985.12 period; the goal is improved implied volatility prediction, so test error for realized volatility is a secondary concern. I present transfer learning and benchmark results below.

Table 12: Transfer Learning and Conventional Model Performance

Config.	Method	Model	Fine-Tuning	Obs.	RMSE	R^2
1	80/20 Split	Gradient Boosting	N/A	58	4.96	0.48
		Multi-Layer Perceptron	N/A	58	5.30	0.35
		Transfer Learning	No	58	1.49	0.72
		Transfer Learning	Yes	58	1.50	0.72
3	Backwards	Support Vector*	N/A	119	7.45	-0.11
		Multi-Layer Perceptron	N/A	119	7.45	-0.12
		Transfer Learning	No	119	2.09	-0.05
		Transfer Learning	Yes	119	2.13	-0.09
4	Forwards	Support Vector	N/A	168	9.62	-0.11
		Multi-Layer Perceptron	N/A	168	12.17	-0.77
		Transfer Learning	No	168	3.07	-0.35
		Transfer Learning	Yes	168	3.53	-0.79

Note: Out of sample measures. * is Manela and Moreira's SVR. Benchmark multi-layer perceptron uses (128,32,8) architecture, Adam optimizer, $\alpha = 0.0001$. Configuration: 1: 1986.01-2009.12 80/20 Random Split on VXO, 2: 1927.07-2009.12 80/20 Random Split on SP500 Realized Vol, 3: Train 1996.01-2009.12 and Test 1986.01-1995.12 on VXO, 4: Train 1986.01-1995.12 and Test 1996.01-2009.12 on VXO. Neural network has three hidden layers (128,32,8) with ReLu activation and output layer (1) with linear activation, mean squared error loss, Adam optimizer, and ExponentialDecay($\alpha = 0.01$, 10000,0.9) learning rate for realized volatility and ExponentialDecay($\alpha = 0.001$, 10000,0.9) for implied volatility, and early stopping (validation MSE, 0.01 tolerance, 5 epoch patience). Network and MLPR trained on count vectorized 5,000 feature document-term matrix. Gradient boosting and support vector regressors use 342,155 features. Realized volatility is 1896.05-1985.12 DJIA, implied volatility is 1986.01-2009.12 VXO.

For each implied volatility training configuration - an 80/20 randomized split of the 1986.01-2009.12 data, training on 1996.01-2009.12 and testing on 1986.01-1995.12, and training on 1986.01-1995.12 and testing on 1996.01-2009.12 as used in steps three and four, I report the top performing conventionally trained model as well as benchmark performance from a multi-layer perceptron regressor with the same architecture as the neural network used in transfer learning. Gradient boosting and the support vector regressions use 342,155 features, whereas the custom neural network and benchmark perceptron use only 5,000. Despite considerably fewer features, using the custom architecture with transfer learning increases out-of-sample R^2 by 0.24 compared to gradient boosting and 0.37 compared to a similar neural architecture for the 80/20 randomized split. RMSE falls from 4.96 to 1.49. The model better predicts the direction in which implied volatility moves and gets closer to the true value. Fine-tuning does not affect performance. For the backwards method, transfer learning has a 0.06 higher R^2 and 5.36 lower RMSE, although R^2 is still marginally negative. Predicting implied volatility in the period 1996.01-2009.12 remains difficult even with transfer learning, although RMSE falls considerably.

I conduct several robustness checks. First, I use a neural network with a constant learning rate of $\alpha = 0.01$ but the same hidden layers (128,32,8) trained on an 80/20 split of the 1986.01-2009.12 implied volatility data and a 5,000 feature count-vectorized document-term matrix. This model achieves out-of-sample RMSE of 7.55 and R^2 of -0.31, confirming my choice of vectorizer and a decaying learning rate. Second, I repeat step one of the transfer learning procedure except using an 80/20 split of realized volatility from 1896.05-1985.12 instead of the 90/10 split used above. RMSE is stable at 1.29 but R^2 falls to 0.65, a result consistent with my earlier test of a multi-layer perceptron and evidence of the network not overfitting to realized volatility. Third, I repeat step one with 100% of the 1896.05-1985.12 sample for training, and use all of the 1986.01-2009.12

VXO series as a test period. Over all 287 months, RMSE is a respectable 4.52 but R^2 is -2.29, rebutting the notion that a model trained exclusively on realized volatility over 1896.05-1985.12 can predict implied volatility from 1986.01-2009.12. Freezing layers preserves the distinction between implied and realized volatility. Fourth, I repeat transfer learning step one as is, retaining the 90/10 split, and predict VXO as in the third check. Doing so tests whether realized volatility, which largely resembles implied volatility from 1986.01-2009.12, has meaningful predictive power for implied volatility and whether the layer freezing component of the procedure is necessary. A R^2 of -0.61 confirms the use of layer freezing.

Table 13: Transfer Learning Robustness Checks

	Realized Volatility			Implied Volatility		
Check	Obs.	RMSE	R^2	Sample	RMSE	R^2
1	-	-	-	58	7.55	-0.31
2	272	1.29	0.65	-	-	-
3	-	-	-	287	4.52	-2.29
4	108	1.08	0.75	287	3.16	-0.61

Note: Out of sample errors. Check 1 is train a neural network has three hidden layers (128,32,8) and $\alpha = 0.01$ on 5,000 feature count vectorized document-term matrix and test on 80/20 split of 1986.01-2009.12 VXO to test choice of vectorizer and learning rate. Check 2 is use custom architecture with early stopping and a decaying learning rate on 1896.05-1985.12 realized volatility but an 80/20 split to test for overfitting to realized volatility. Check 3 is use architecture form check 2 but no test data for realized volatility, and test on 1986.01-2009.12 VXO. Check 4 is check 3 but with a 90/10 realized volatility split and predicting 1986.01-2009.12 VXO without transfer learning.

Improvements in R^2 are noteworthy but the reduction in RMSE and use of considerably fewer features is critical. Exposing a model to the events and realized volatility of the period 1896.05-1985.12 reduces the number of text features needed from 342,155 to a mere 5,000. A model's knowledge of a textually less descriptive but longer history lets it perform better than a model with a textually richer knowledge of a shorter history.

8.1 Transfer Learning Variance Risk Premium

Fine-tuning involves unfreezing all layers of a neural network and retraining the model at a very low learning rate. This results in small adjustments to the model's weights while preserving feature representations learned earlier. I use fine-tuning as a last transfer learning step, despite the decrement to final out-of-sample performance over the 1986.01-2009.12 period, to ensure the neural network retains the distinction between implied and realized volatility. Re-exposure to the relation between textual features and implied volatility with trainable preliminary layers pushes the model closer to predicting implied volatility than realized volatility. To measure the effect of fine-tuning, I compute the estimated variance risk premium for a series generated with and without fine-tuning, and compare it again to the measured variance risk premium over the 1990.01-2021.12 period.

Table 14: Transfer Learning Estimated Variance Risk Premium

	Fine-Tuning 1896.05-2009.12	No Fine-Tuning 1896.05-2009.12	Measured VRP 1990.01-2021.12
Count	1359	1359	384
Mean	7.31	8.01	15.10
Std.	32.16	28.23	29.66
Min.	-511.49	-514.29	-403.40
p25	3.02	5.05	7.47
p50	12.03	10.98	12.85
p75	19.46	17.43	22.73
Max.	128.84	128.62	115.85
$\rho_{IV,RV}$	0.85	0.89	0.78

Note: $\widehat{\mathrm{VRP}}_t = \mathrm{IV}_t - \mathrm{RV}_t$ for $IV_t = \mathrm{VIX}^2/12$ and realized variance equals the sum of squared log returns. Over 1990.01-2021.12, the variance risk premium is measured and reported (Zhou 2018).

Without fine-tuning, estimated option implied variance has a correlation of 0.89 with realized variance. After fine-tuning, the correlation falls to 0.85. The former seems atypi-

cally high compared to the observed 0.78. A lower correlation confirms that incorporating fine-tuning results in an estimated implied volatility more dissimilar to realized volatility. The minimum, maximum, and standard deviation figures are in line with the observed period but averages with and without fine-tuning, 7.31 and 8.01 respectively, are lower than the observed 15.10. Investors possibly were less averse to risk or had a shallower understanding of stochastic volatility during the 1896.05-1989.12 period. Evidence for the latter is the implied volatility smile, which arose only in the aftermath of the 1987 crash. This indicates that investors didn't fully comprehend stochastic volatility pre-1986.

I plot the fine-tuning estimated implied volatility series. As expected, implied volatility is on average higher than realized volatility, but during key crises, such as the Crash of 1929, outbreak of WWI, and Pearl Harbor, realized volatility is higher.

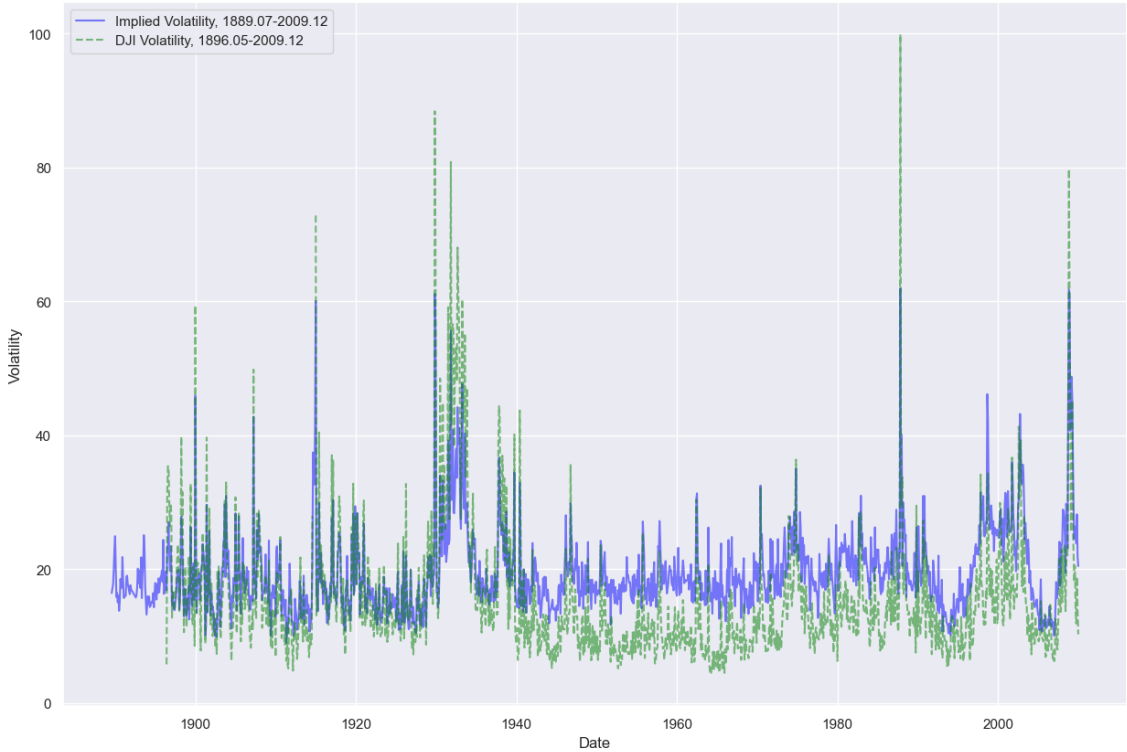


Figure 5: Implied Volatility Estimated from Transfer Learning and DJIA Realized Volatility, 1889.07-2009.12

9 Discussion

Coherence with established styled facts regarding the variance risk premium is one empirical test for the plausibility of estimated historical implied volatility. The best models trained and tested on implied volatility data from 1986.01-2009.12, such as this paper's gradient boosting regression and Manela and Moreira's support vector regression, produce an estimated variance risk premium that is positive on average but for which option implied variance has insufficient correlation with realized variance and for which the variance risk premium is not negative in extreme crises. It is entirely possible that the period in which the variance risk premium is measured, 1990.01-2021.12, constitutes a structural break from the prior period, but this criticism is applicable to any empirical test because it necessarily involves some comparison of implied volatility data post-1986 with macroeconomic variables. It suffices to say that correlations between option implied and realized variance of 0.47 and 0.89 both strain credulity. The reduction to 0.85 with fine-tuning is definitive proof that a fine-tuning reinforces the model's understanding of the distinction between implied and realized volatility.

A reductive spot test for estimated historical implied volatility is that it resembles realized volatility, but tends to be smoother, on account of being a forward-looking measure, and higher, on account of investors' risk aversion. The series estimated from transfer learning with fine-tuning passes with flying colors and obviates any creative re-interpretations of major historical events. Large deviations in implied volatility accord with the historical record on the relative magnitude of shocks[42]. For example, the Panic of 1907 was the first global financial crises of that century, dwarfing the Panics of 1893 and 1896, and this is reflected in higher estimated implied volatility. The levels are more appropriate as well,

[42]Manela and Moreira's estimates of implied volatility during the Panic of 1901 are higher than any point in US history until the Russian default and LTCM blowup of 1998. This is wildly implausible.

with implied volatility subsiding during the roaring 20s and economic expansion of the 1950s, but remaining elevated during the Great Depression and the inflationary period of the 1970s and early 1980s. The problem with smoothing realized volatility and adding non-negative noise to get a visually plausible estimate of implied volatility is that the gap between realized and implied volatility is no longer time-varying in a principled manner.

Given the discussion of the variance risk premium, one may wonder about the feasibility of calibrating a time series model of the variance risk premium to the 1990.01-2021.12 period, or developing a model of implied volatility as a function of observable macroeconomic covariates like realized volatility, employment, and production. I eschew this approach for the following reasons. First, the set of covariates is limited to those available from around 1900, such as industrial production, employment, savings and investment, and the price level. Preliminary analysis showed these have little power to predict implied volatility contemporaneously, so predicting over a long out-of-sample period was dubious. Exacerbating this issue is the quarterly or annual frequency for many key time series. Second, these models suffer the same deficit of using relationships established after 1986 to estimate the relation between implied volatility and the same set of covariates decades prior. Third, there is an obvious endogeneity problem. Investors consume macroeconomic data to update their expectations, which manifests in options market activity that drives levels of forward-looking implied volatility up or down. Investors, firms, and consumers also change their economic behavior in response to levels of uncertainty, which manifests in updated macroeconomic data. Both channels are forward-looking, so any such model obliterates the distinction between these mechanisms when predicting backwards. Exogenous shocks are passed to investors information sets through news coverage but implied volatility itself does not drive news coverage, favoring my approach. Fourth, and most importantly, the point of this paper's exercise is to capture the spirit of the times, mean-

ing a broad measure of sentiment, uncertainty, and outlook, at various points in history. Hard data simply cannot reflect this in the way that news coverage can. On August 5th, 1914, the headline of the Journal read "GERMANY WARS ON FRANCE; ENGLAND STANDS BY BELGIUM". Industrial production figures did not reflect US involvement in World War I until years later, let alone the calamity of the moment.

I discussed the challenges of using a randomized train-test split approach and document evidence of optimistic test performance for particular random train-test splits. Based upon my non-parametric bootstrap analysis, which showed stable test error for most data splits, I proceeded to use a randomized split in the transfer learning procedure. That out-of-sample R^2 improves markedly when predicting VXO during step 4 of the procedure, regardless of the particular implied volatility or realized volatility split, attests to the utility of transfer learning. The operative question is how much R^2 and RMSE improve when using transfer learning, not the absolute model performance. I varied the random split of 1986.01-2009.12 VXO used to train a gradient boosting regressor conventionally and to perform transfer learning, using the same split for both, and identified consistent increases in R^2 on the order of 0.20-0.25. Transfer learning exposes the model to a broad swath of pre-1985 historical events, some of which bear resemblance to events in the most difficult periods to predict post-1986 and thus improves predictive power.

Interpretability is an often cited concern for neural networks. With several hidden layers, hundreds of neurons, non-linear activation functions, and millions of parameters, the meaning of any individual text feature cannot be identified in the way a covariate in a linear regression can. Interpretability is second to prediction in this exercise, but the unsupervised topic model analysis illustrates that WSJ text does capture meaningful events and the persistence of their news coverage over time. Combined with results from more interpretable models like penalized regression, this should allay such concerns.

10 Conclusion

Decades after key historical events in the late 19th and 20th centuries, the spirit of bygone eras - fear, uncertainty, euphoria - is lost in hard data. Employment or industrial production figures do not capture the concerns or expectations of average investors. Since 1986, the Chicago Board Options Exchange has published an index that use options prices to measure expected volatility on the S$P100 and then S&P500 indices. VIX today is a preeminent gauge of equity market volatility and investor sentiment, but is not reported prior to 1986 owing to a lack of data. I answer the question of what implied volatility, a market based measure of the spirit of the times, could have been during seminal historical periods, like the Panic of 1907, the Crash of 1929, the outbreak of World War II, or the Cuban Missile Crisis of 1962. Leveraging the rich historical textual news coverage from the Wall Street Journal dating to July 1889, I estimate historical implied volatility using a novel transfer learning approach. At first glance, predicting option implied volatility from Wall Street Journal articles appears to be an exercise in futility. Implied volatility itself is a forward-looking market measure of uncertainty derived from options prices, and textual news coverage merely reflects events that transpired. Using exactly that news coverage, natural language processing, and techniques from high-dimensional statistical inference, I show that historical implied volatility can be reliably estimated over a near hundred year span in which options price data is not available.

Furthermore, I document the performance of a range of high-dimensional statistical models, data reduction techniques, and training and testing methodologies for this task, and validate my estimates with key stylized facts regarding the variance risk premium. Using a very modest neural network, I demonstrate that transfer learning from realized volatility to implied volatility outperforms conventional training methods while decreasing

data used. Models training using data from January 1986 to December 2009 perform poorly because events pre-1986 are quantitatively and qualitatively different. Transfer learning results in a model that can better predict the direction implied volatility moves, evident by higher R^2, and more accurately predict the level of implied volatility, evident by lower root mean squared error. A more complicated network structure and natural language processing techniques, such as pre-trained word embeddings to better capture semantic relationships and context, may perform even better. Logical next steps include accessing the raw Wall Street Journal text data and expanding the period of implied volatility and text data from December 2009 to present day. I discussed the deficiencies of various time series approach, but follow-up analysis would include benchmarking various models against the series estimated in this paper. A vector autoregression involving realized volatility, implied volatility, and macroeconomic covariates could be fit to the post-1986 period and to the pre-1986 period estimated series to compare the stability of coefficients. Alternatively, I could simulate from the VAR using historical macroeconomic data and coefficients estimated from the post-1986 period to get an estimate of implied volatility pre-1986. The VAR could incorporate structural event-driven restrictions as done in Ludvigson et al (2021).

This paper produces an empirically grounded implied volatility series, spanning July 1889 to December 2009, that matches the historical record and quantifies uncertainty and market expectations of volatility from decades ago. This series augments the historical record and enhances understanding of financial history, in addition to providing a new dataset with many application to empirical asset pricing.

11 Appendix

11.1 Historical Realized Volatility

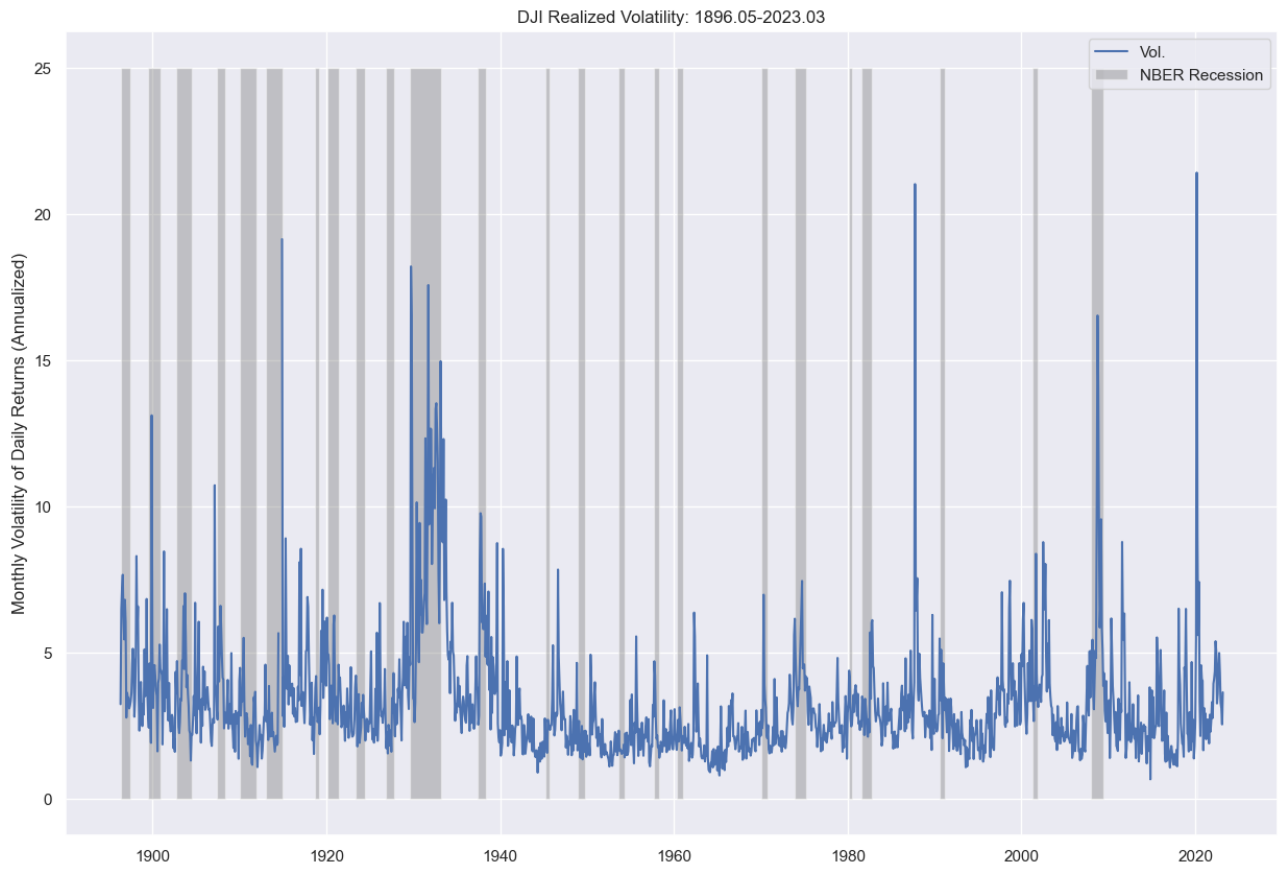


Figure 6: Dow Jones Index Volatility, 1986.05-2023.03

In 1884, Charles Dow published a stock index comprised of 11 companies (9 railroads, 2 industrial companies), reflecting the dominance of railroad companies in the US economy at the time. On May 26th, 1896, he published the precursor to the Dow Jones Industrial Average consisting of 12 industrial stocks. Original companies included American Tobacco, American Sugar Refining, US Leather, US Rubber, and General Electric. This series has been reported daily ever since, except between July 30th and December 12th,

1914 as a result of the outbreak of WWI. From daily index values I obtain daily returns and a monthly variance or volatility of those returns. Volatility spikes during each financial panic in the early 1900s, stock market crashes in 1929, 1987, and 2008, the outbreak of WWI, WWII, the Korean War, the Cuban Missile Crisis, and 9/11.

11.2 Realized Variance, Daily vs. Intra-day Returns

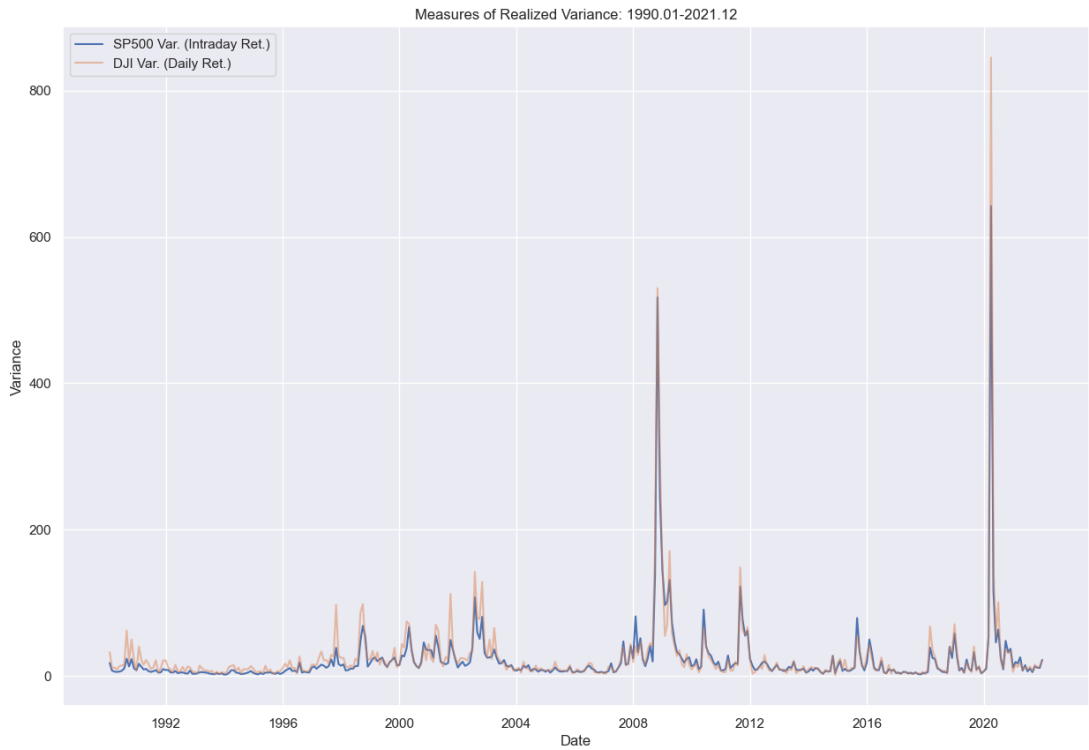


Figure 7: Realized Variance, 1990.01-2021.12

I plot my measure of realized variance using DJIA daily returns against a measure of realized variance using SP500 intra-day returns from a recent paper on the variance risk premium (Zhou 2018). The series have a 0.98 correlation.

11.3 Reconstructing the WSJ Corpus

Manela and Moreira use the contents of the front-page of the Wall Street Journal from July 1889 to December 2009 as a corpus to perform text analysis. Because I do not have access to the raw text data, I rely on the replication kit for the paper to reconstruct the document-term matrix used in their analysis. Lack of data access prevents me from extending the training or test periods forward past 2009.

The authors publish three .txt files, one for each of the train, test, and predict subsamples. The train sample spans the period of January 1996 to December 2009, the test sample, January 1986 to December 1995, and the predict sample July 1889 to December 1985. Each line in these .txt files is a row in a document-term matrix and corresponds to a month's worth of text data, and also includes the data and month-end VXO value. For the predict file, the VXO value is recorded as 0. They also supply a .txt file that maps the corpus' vocabulary to feature identification numbers. The vocabulary contains 468,091 n-grams and a visual inspection shows predominantly unigrams and bigrams.

Reconstructing the document-term matrix gives me flexibility to alter some of the text processing steps the authors took, despite not having the raw text data. For example, I noticed the authors use a weak form of stemming when creating a corpus wide vocabulary. The result is countless n-grams which are fundamentally the same but noted as separate features in the author's document-term matrix, such as nice, nicer, nicely, and nicest. I restrict the feature space in part by stemming the vocabulary and combining the counts of features with the same stemmed form. The paper's vocabulary includes n-grams with leading or trailing underscore characters. For example, Table 7 (Manela and Moreira 147) shows Stock, _ Stock, U.S., and _ U.S. as different n-grams with distinct counts. The stated purpose is to replace stopwords in n-grams with an underscore. My view is that the effect is to overstate the count of the underlying unigram. Stopwords should be

removed and then n-grams computed, so I further restrict the feature space by removing these trailing or leading underscore characters.

With a smaller document-term matrix (fewer text features), I can examine changes in the performance of the text regression. The paper describes x_t as a vector of dimension 1×468091 wherein:

$$x_{t,i} = \frac{\text{appearances of n-gram i in month t}}{\text{total n-grams in month t}}$$

meaning the n-gram count is scaled by the total number of unique n-grams in a given month. However, the values seem like raw counts *multiplied* by a scaling factor. For exxample, a count value of 4 is scaled by 77.429 and reported as 309.71738. I divided each element by the smallest element to get the original raw counts. Despite a vocabulary of size 468091 and 1,439 months worth of text data, the actual document-term matrix is quite sparse. Only 1.5% of all elements in the matrix are non-zero, indicating that a parsimonious document-term matrix can be used to produce similar results and possibly less over-fitting to training data. Another benefit is reduction in computational cost. I examine the effects of data reduction on model performance.

11.4 Vocabulary Reduction

Vocabulary refers to the set of all unique n-grams in a corpus computed after all pre-processing steps, such as removal of stopwords, punctuation, numeric characters, and stemming. In a document-term matrix, each column represents the occurrences of a single element of the vocabulary across all documents in the corpus, and a feature to be used in predicting VIX. Trimming the vocabulary is hence a model agnostic feature selection method. The original dictionary contains 468,091 n-grams, the majority of which are unigrams.

Table 15: Vocabulary Reduction

Approach	Size	Features Eliminated	Percent of Original
Original	468,091	0	100.0
Remove _	400,080	68,011	85.47
Remove Stopwords	399,951	68,140	85.44
Remove n-gram size 1	342,179	125,912	73.10
Combined Stemmed Forms	342,155	125,936	73.09
Parsimonious	5000	463,091	1.07

Each n-gram in the first 30,000 entries of the vocabulary is an n-gram that appears subsequently in the vocabulary but with an underscore appended or inserted. Regardless of the rationale, such as a placeholder for a numeral, or stand in for variants of the same n-gram (e.g. stock _ represents stock sale, stock broker, stock fell, etc.), these n-grams clutter the feature space and add diminishing value. The constitutive unigrams already exist elsewhere in the vocabulary, often at much greater frequency, and the n-grams with an underscore add no value with respect to intepretability[43] . Eliminating such n-grams reduces the vocabulary by 68,011 n-grams. A further 129 n-grams are in the NLTK English stopwords list, so I remove those[44]. Next I stem each n-gram and combine the base forms. For example, flying, flys, and flyer are all reduced to fly. Combined with removing n-grams of length 1, the vocabulary is reduced by 125,936, or 27.11 %. Again, the sparsity of the original document-term matrix motivates this effort. I document the performance of various regression models on the reduced feature set.

[43]It is possible to retain some token dependence in the bag-of-words representation by use of n-grams. N-grams are phrases of length n and for $n \geq 2$ permit a richer representation of text (footnote: stocks fell, stocks rose, better than stocks, fell, rose, individually). The tradeoff with use of n-grams is increase the dimension of the fixed-length vectors representing a document, and beyond $n = 3$ n-grams exhibit diminishing marginal returns. Regardless, the vocabulary set in Manela and Moreira's replication kit considers at most trigrams.

[44]The list of stopwords can be manually or systematically augmented. For the former, a logical extension is contextual stopwords, which are words that convey little unique meaning in a specific corpus but would otherwise be important, unlike simple articles or conjunctions. For example, in a physical sciences corpus, the words "chemistry", "biology", "chemist", are uninformative given the context. For the latter, consider excluding words with counts or TFIDF scores below a particular threshold.

I finally take a 5,000 most frequently occurring word subset of the vocabulary for testing. The size is arbitrary but represents the limit of data reduction, throwing away the vast majority of the document-term matrix. This results in three document-term matrices to test the effects of data reduction. I perform a TFIDF transformation of the count document-term matrices as well for a total of six matrices.

Without the original corpus, I cannot add tokens back to the vocabulary, such as n-grams containing digits (footnote: For example, "9/11" or "September 11th") that were removed by Manela and Moreira. Without raw text, the more complicated text representations, like word embeddings, are difficult to implement.

I further reduce the feature set by stemming tokens and retaining only the root form[45]

[45]The tokens 'President', 'President's', 'Presidential', are replaced with President.

11.5 Text Similarity Regression Robustness Checks

The positive coefficient for the linear-linear model with an intercept indicates that textually identical months have an absolute VXO difference of $\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 = 11.49 - 10.32 = 1.17$.

Table 16: Linear-Linear Text Similarity Regression

	$\lvert \text{VXO}_i - \text{VXO}_j \rvert = \beta_0 + \beta_1 \cdot \cos(\theta_{ij}) + \epsilon_{ij}$	
	No Intercept	Intercept
Cosine Similarity	5.68***	-10.32***
	(14.02)	(-6.18)
Constant		11.49***
		(9.79)
Observations	287	287
R^2	0.41	0.12
Note:	*p<0.1; **p<0.05; ***p<0.01. T-stat in parentheses.	

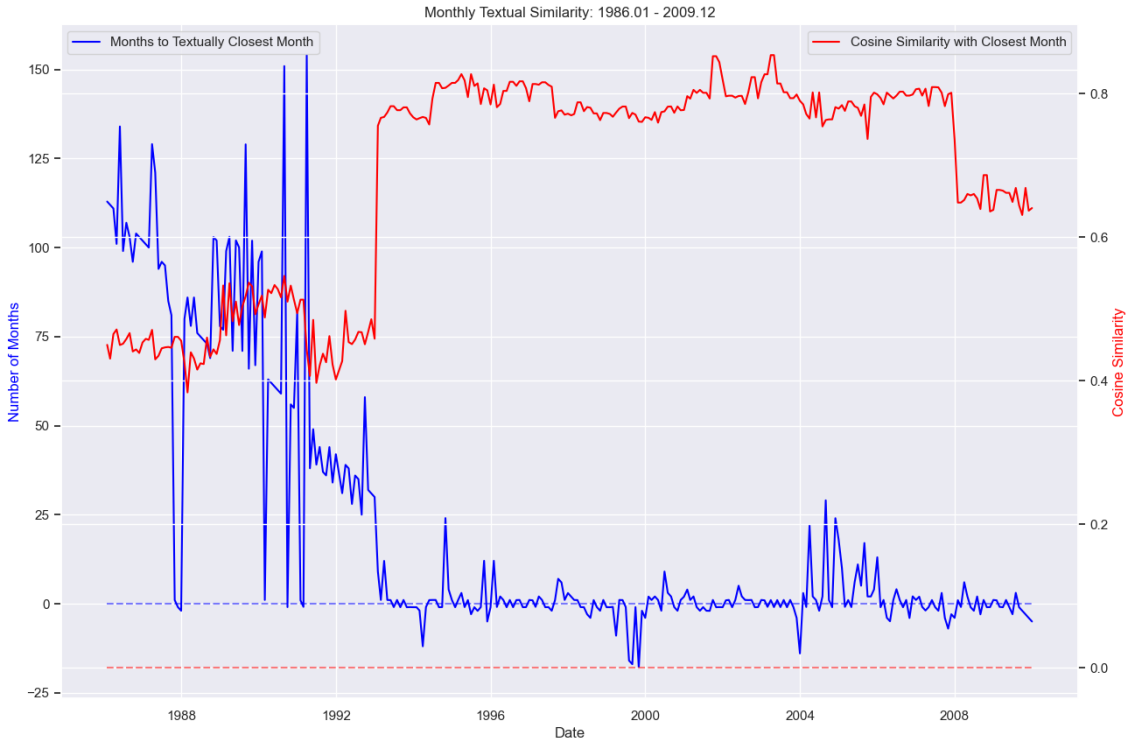
Table 17: Log-Log Text Similarity Regression

	$\ln\lvert \text{VXO}_i - \text{VXO}_j \rvert = \beta_0 + \beta_1 \cdot \ln(cos(\theta_{ij})) + \epsilon_{ij}$	
	No Intercept	Intercept
Log Cosine Similarity	-2.05***	-1.35***
	(-13.57)	(-4.65)
Constant		0.38***
		(2.81)
Observations	287	287
R^2	0.39	0.07
Note:	*p<0.1; **p<0.05; ***p<0.01. T-stat in parentheses.	

A 1% increase in textual similarity decreases absolute VXO difference by 1.35%.

11.6 Persistence of News Coverage

A concern with this test of textual similarity is that the regression is simply picking up the persistence of news coverage around large shocks that drive elevated implied volatility. Concerns over adverse shocks persist month to month, so news coverage in sequential months and implied volatility happen to be similar. To examine this, I plot the number of months between the target month, on the X-axis, and its textually closest (by cosine similarity) month.



The difference is closest month minus target month, so a positive value on the blue line means the closest month is Y months ahead of target month, and a negative value means closest textual month is prior to the target month. From 1986 to 1993, news coverage becomes less varied as news text increasingly centers on the same topics. From 1993 forward, the cloest month is often the month prior or after. Spikes above and below are

akin to attention shifts and they subside shortly. The red line shows cosine similarity and is markedly higher after 1993. I partition the sample at 1993 and test each subsample:

Table 18: Linear-Log Text Similarity Regression, Subsamples

	$\lvert \text{VXO}_i - \text{VXO}_j \rvert = \beta_0 + \beta_1 \cdot \ln(cos(\theta_{ij})) + \epsilon_{ij}$	
	1986.01-1993.01	1993.02-2009.12
Log Cosine Similarity	-6.06	-12.51***
	(-0.887)	(-3.13)
Constant	1.90	0.33
	(0.36)	(0.32)
Observations	83	204
R^2	0.01	0.05
Adjusted R^2	-0.00	0.04
Residual Std. Error	5.01	3.79
F Statistic	0.79	9.79***
Note:	*p<0.1; **p<0.05; ***p<0.01. T-stat in parentheses.	

and find the negative relationship between text similarity and difference in implied volatility holds up from February 1993 to December 2009 but not the prior period. Indeed, text similarity regression results do appear to be driven by the persistence of news coverage. I argue however this is consistent with the hypothesis that news text proxies investors concerns and thus market measure of uncertainty. That months with more similar news coverage have similar levels of implied volatility after January 1993 but not prior simply indicates a noisy relationship prior. The large deviations in number of months between target month and closest month are accompanied by middling cosine similarity, showing that the pair of months have insignificant textual similarity and thus one shouldn't expect similar VXO. The hypothesis that WSJ news text encapsulates the concerns of forward-looking investors is entirely plausible.

11.7 Implied Volatility and Latent Topics

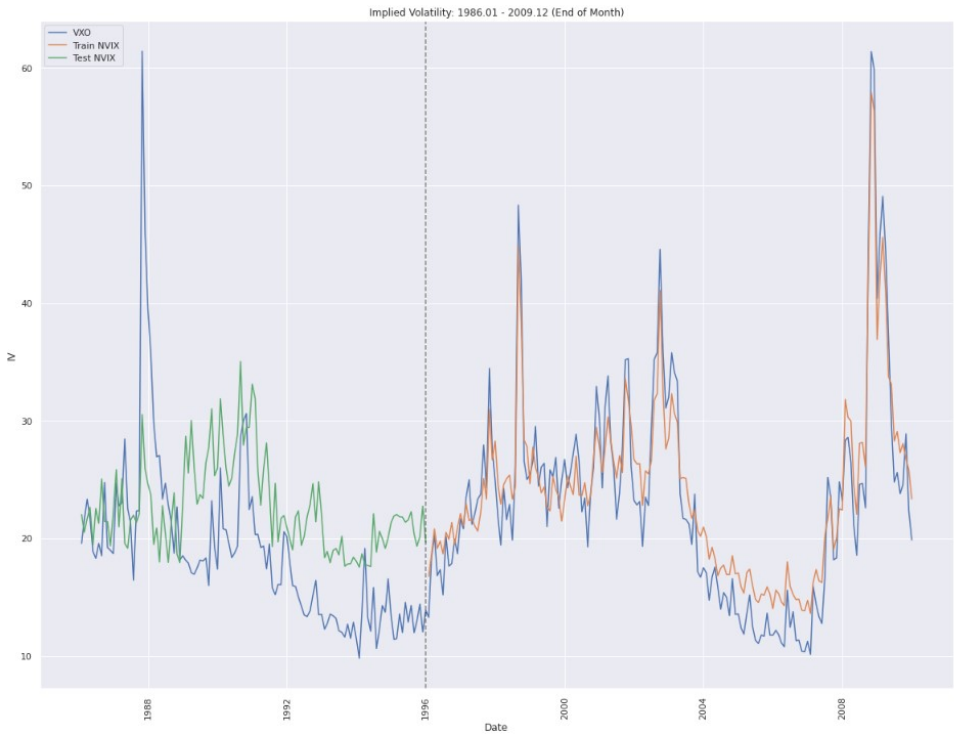
Table 19: VXO Exposure to Latent Topics, 1986.01-2009.12

	$\text{VXO}_t = \sum_{i=1}^{\text{Num. Topics}} \beta_i \text{Topic i}_t + \epsilon_t$						
Topic 1	0.49***	0.64***	0.67***	0.66***	0.66***	0.62***	0.57***
	(0.18)	(0.17)	(0.16)	(0.16)	(0.16)	(0.14)	(0.14)
Topic 2		-0.91***	-0.91***	-0.91***	-0.91***	-0.92***	-0.91***
		(0.15)	(0.14)	(0.14)	(0.14)	(0.12)	(0.12)
Topic 3			-0.56***	-0.60***	-0.60***	-0.77***	-0.79***
			(0.13)	(0.13)	(0.13)	(0.11)	(0.11)
Topic 4				1.25***	1.25***	1.30***	1.09***
				(0.42)	(0.42)	(0.37)	(0.36)
Topic 5					-0.01	-0.04	-0.03
					(0.10)	(0.09)	(0.09)
Topic 6						1.80***	1.74***
						(0.19)	(0.19)
Topic 7							0.83***
							(0.21)
Observations	287	287	287	287	287	287	287
R^2	0.03	0.14	0.20	0.22	0.22	0.41	0.44
Adjusted R^2	0.02	0.14	0.19	0.21	0.21	0.40	0.43
Residual Std. Error	0.99	0.93	0.90	0.89	0.89	0.78	0.76
F Statistic	7.81***	23.88***	23.34***	20.21***	16.11***	32.60***	31.71***

Note: *p<0.1; **p<0.05; ***p<0.01
No intercept, dependent variable is Z-scored. Standard errors in parentheses.

I regress VXO on the monthly topic distributions, using the LDA model fit to the 1986.01-2009.12 data and 7 topics. No intercept is included to avoid collinearity (the covariates sum to 1 by construction). All topics are significant except for Topic 5, confirming that VIX is reponsive to changes in the topics of news coverage. VXO is Z-scored, so a 0.1 increase in topic probability i coupled with a 0.1 decrease in topic probability j results in a $0.1\hat{\beta}_i - 0.1\hat{\beta}_j$ standard deviation change in VXO. Because the covariates are probabilities, counterfactual analysis that increments one probability must decrement another or several other to ensure probabilities still sum to 1. Topic 7 is terrorism and 1 is politics, so a 0.1 increase in topic 7 probability and 0.1 decrease in Topic 1 probability raises VXO by 0.026 standard deviations.

11.8 NVIX Deficiencies



An obvious question is why improve on an estimate of historical implied volatility that was already published in the Journal of Financial Economics. In this section, I document many problems with the work of Manela and Moreira. The paper's replication file contains the predicted NVIX series from 1889 to 2016 and observed VXO from 1986 to 2009. I first examine the true out-of-sample performance of the author's SVR using the test period of 1986.01 - 1995.12. I also examine in sample performance for evidence of overfitting. The authors report out of sample R^2 of 18.53 and RMSE of 7.48 for 119 observations (they cite lack of data for January, 1992). For the regression of predicted NVIX on observed, they identify coefficient estimates of 0.17 and 0.82, with standard errors of 4.37 and 0.20, respectively, and R^2 of 19.46.

I document out of sample RMSE of 7.45, MSE of 55.55, MAPE of 0.3669, author's

R^2 of 0.1808, and conventional R^2 of -0.11. Their measure of R^2 is wrong. See the section on regression test error for more detail on the source of the discrepancy. For the contemporaneous regression of predicted on observed, I obtain estimates of 0.36 and 0.815 with standard errors of 4.386 and 0.206, respectively, and R^2 of 0.1906. For a null of $\beta = 0$, the t-stat is 3.96 and for a null of $\beta = 1$, the t-stat is -0.89, so the estimate is statistically greater than 0 but not different than 1. These are in line with the figures reported in the paper. The MAPE and true R^2 is striking. On average, the SVR's predicted VXO deviates from observed by 36 percent and the model performs worse than a baseline of predicting the mean of observed VXO.

Table 20: Paper NVIX Performance

Num. Obs Sample	RMSE	MAPE	Paper R2	True R2	a SE(a)	b SE(b)	Reg. R2
168 IS	2.833	0.132	0.912	0.904	-5.75 0.433	1.204 0.019	0.94
119 OOS	7.45	0.367	0.181	-0.111	0.360 4.386	0.815 0.206	0.191

The above figures are computed using the NVIX series reported in the replication kit, not my replication of their SVR model. To check their SVR model's performance on realized volatility, I fit an SVR model using the exact same training methodology and the reported parameters $C_{SVR} = 50.7369, \epsilon_{SVR} = 3.486$.

Compared to the results in Table 2 of the paper, I document lower train RMSE of 3.017 compared to 3.38, but considerably higher test and predict sample RMSE of 16.45 and 19.615, respectively, compared to reported 9.61 and 10.68. The R^2 from OLS regressions of actual realized volatility on news implied volatility are also lower at 0.132 and 0.0811 respectively, compared to 0.2035 and 0.1599 in the paper. My predict subsample has

Table 21: Paper SVR Realized Volatility Performance

Num. Obs Sample	RMSE	MAPE	Paper R2	True R2	a SE(a)	b SE(b)	Reg. R2
168 Train	3.017	0.199	0.934	0.918	-4.62 0.462	1.172 0.024	0.954
119 Test	16.456	1.525	-0.095	-3.613	0.439 2.134	0.433 0.103	0.132
702 Predict	19.616	2.112	0.069	-3.376	-37.29 7.072	1.649 0.237	0.081

considerably fewer observations, 702 vs the 1150 in the paper, but the test subsamples are the same. However, the ratios of the RMSEs of the test and predict subsamples are not markedly different. The predict sample RMSE is 19.19 % greater in my replication compared to 11.13 % in the paper.

11.9 Return Predictability

I test the ability of the SVR NVIX and the GBR NVIX to predict returns. I use monthly market excess returns and lagged NVIX with Newey-West standard errors. The number of lags equals the return forecasting window. The specification is:

$$R^e_{t,t+\tau} = \alpha + \beta \cdot NVIX_{t-1} + \epsilon_{t+\tau}$$

Note, I do not report standard errors that account for the dependent variable itself being an estimate. I use the same periods in the paper and forecasting windows as the paper. To be concise, I report regressions for which the estimate of β is statistically significant at the 5% level. I find that only the SVR NVIX has predictive power for the periods 1945.01 - 1995.12 and 1945.01 - 2009.12. The GBR NVIX is not a statistically significant predictor of future returns for any forecasting window or time period.

Table 22: NVIX Return Predictability: 1945.01-2009.12

	$R^e_{t,t+\tau} = \alpha + \beta \cdot NVIX_{t-1} + \epsilon_{t+\tau}$	
	$\tau = 3$	$\tau = 6$
NVIX_{t-1}	0.019***	0.01**
	(3.49)	(1.97)
Constant	-0.36***	-0.16
	(-2.89)	(-1.38)
Observations	780	780
R^2	0.02	0.01

Note: *p<0.1; **p<0.05; ***p<0.01
Newey-West standard errors. T-stat in parentheses.

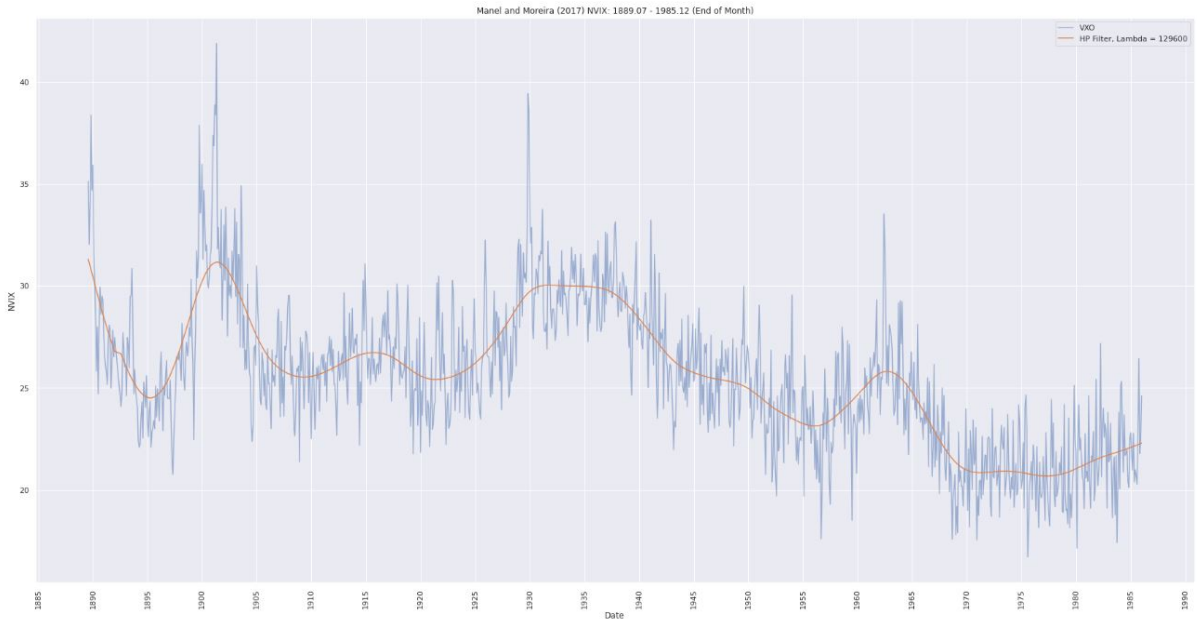
My results are very different from those in the paper. The paper finds a statistically significant relationship for the 1945-1995 period at all forecasting windows and for windows 6, 12, and 24 for 1945-2009. However, the coefficients reported in the paper are much larger and R^2 is much higher, ranging from 0.15 to 0.37 and 0.35% to 5.02% respectively. GBR NVIX has no predictive power for monthly returns.

Table 23: NVIX Return Predictability: 1945.01-1995.12

	$R^e_{t,t+\tau} = \alpha + \beta \cdot NVIX_{t-1} + \epsilon_{t+\tau}$		
	$\tau = 1$	$\tau = 3$	$\tau = 6$
NVIX_{t-1}	0.018**	0.031***	0.017**
	(2.29)	(4.62)	(2.15)
Constant	-0.34*	-0.65***	-0.31*
	(-1.83)	(-3.96)	(-1.67)
Observations	612	612	612
R^2	0.01	0.03	0.01

Note: *p<0.1; **p<0.05; ***p<0.01
Newey-West standard errors. T-stat in parentheses.

11.10 NVIX Historical Plausibility



Instead of examining the plausibility of Manela and Moreira's NVIX with specific events, for which the fit is simply too noisy, I apply the Hodrick-Prescott filter to the NVIX time series and obtain a smoother trend to compare levels of NVIX over time periods.

The period of 1900 to 1910 shows declining NVIX, despite the Panic of 1907 from May 1907 - June 1908. The event cited in the paper, the Panic of 1901, is regarded historically as smaller and less disruptive. The Panic of 1907 was the first global financial crises of the 1900s and triggered a deep recession and the creation of the Federal Reserve, yet NVIX indicates a much larger shock from the prior panic. NVIX from 1910 to 1915 exhibits a mild upward trend and seems to understate disaster risk in the lead up to World War I. NVIX actually falls in 1914 at the outbreak of World War I, which was triggered by an assassination. The series rises during the 1920s which seems incorrect.

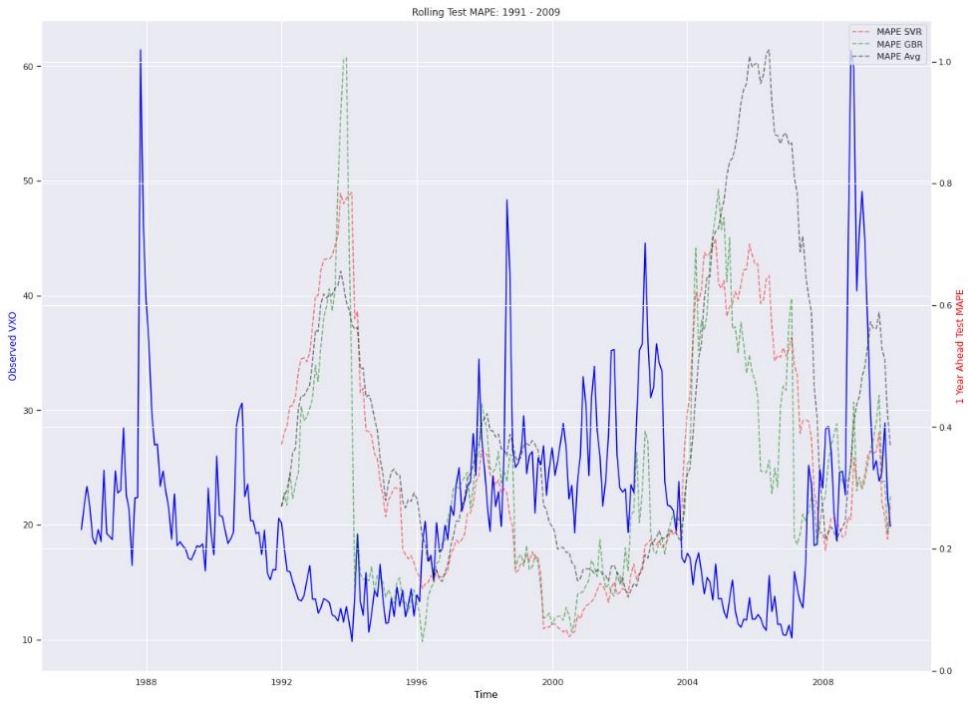
Known as the "Roaring Twenties", the decade was a period of broad economic recovery and growth, suggesting NVIX should fall. NVIX does capture the market crashes in October and November 1929, however the levels during the 1930s are higher than the preceding or subsequent decade despite the entrance of the US into WWII after Dec. 7th, 1941. NVIX during WWI is modestly hump shaped but declining during WWII. There were two recessions in the US in the second half of the 1940s, one from Feb.-Oct. 1945 and another from Nov. 1948 - Oct 1949, but otherwise the period was comparatively uneventful, so the decline in NVIX seems reasonable. NVIX completely misses the outbreak of the Korean War in June 1950 but the rise between 1955 and 1960 roughly coincides with the US recession from Aug. 1957 - April 1958. During the US recession from April 1960 - Feb. 1961, NVIX falls. NVIX seemingly mistimes the Cuban Missile Crisis of Oct.-Nov. 1962, showing a modest increase in May 1962, but the US public was only aware of Soviet activities in Cuba by August, 1962. This event should register in NVIX as a considerably greater disaster risk but does not. That the event is such a historical anomaly explains why text features associated with it do not show up in the training set. The downward trend in NVIX from 1965 to 1970 is without historical explanation. 1968 in particular was a tumultuous year, with the Tet Offensive, Prague Spring, countless riots and protests, and assassinations. Most puzzlingly, NVIX is flat during the 1970s, despite the US leaving the gold standard, the devastating OPEC oil embargo of Oct. 1973, and acceleration of a price-wage spiral from 1973 to 1975 in the US. NVIX trends upward from 1980 onwards, coinciding with a US recession in the beginning of 1980 and another from July 1981 - Nov 1982 on account of actions by the Federal Reserve. Ultimately, I find the author's lack of explanation for their NVIX series being unresponsive to known historical shocks or disproportionately responsive to comparatively milder shocks to be more concerning than the interpretations of unexpected NVIX spikes, such as the authors citing President

Eisenhower's budget in the early 1950s.

Regardless, the trend lines however are much easier to reconcile. NVIX should be rising from 1965 - 1970, for example, and should not be higher from 1900 - 1905 compared to 1914-1918 or 1941-1945. One explanation for the deficiency of NVIX is that the extreme shocks of the 20th century are simply not present in the training period of 1996 - 2009, so a model trained on text data from this period won't have text samples corresponding to world wars or oil embargoes, for example.

11.11 Rolling Training Window

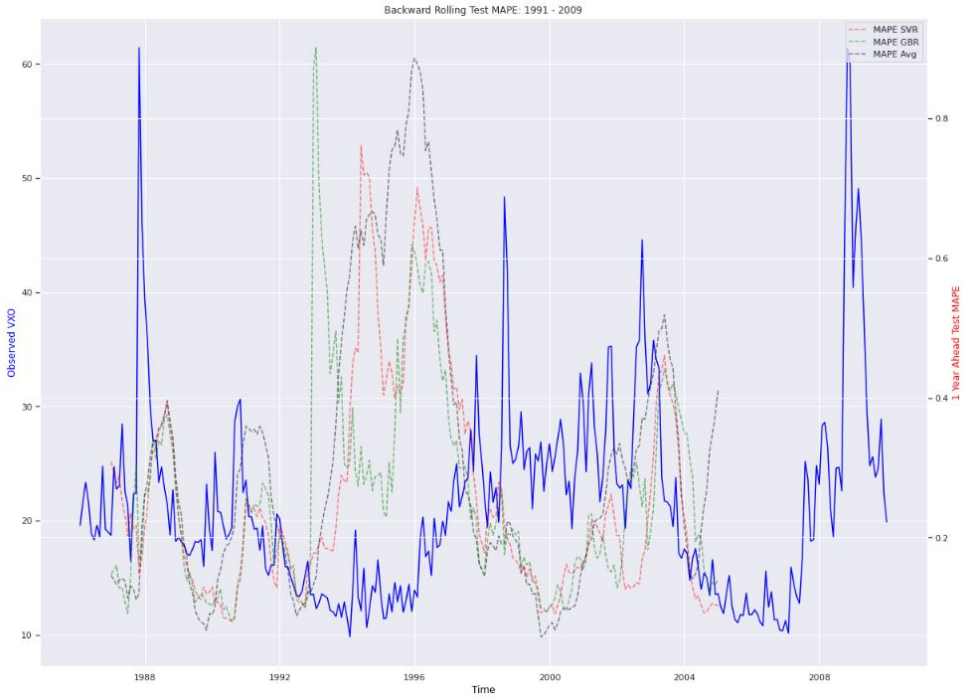
To isolate the effect of training on one period of time and testing on the subsequent period, I use a rolling training strategy in which I train on a particular window of 5 years, predict the next year's worth of observed VXO, and compute performance metrics. I then slide the window across the range of years.



The above plot shows MAPE for a support vector regression model trained on 5 years worth of monthly text data and tested on the next year's worth of monthly text data. For example, the first pass trains on data from 1986.01 - 1990.12 and tests on 1991.01 - 1991.12. The second trains on data from 1986.02 - 1991.01 and tests on 1991.02 - 1992.01, and so forth a total of 216 times. The last training sample is 2004.01 - 2008.12 and the last test sample is 2009.01 - 2009.12. For comparison, I train a gradient boosting regression model in the same manner and plot its MAPE. I add a constant model which predicts the the training period average for reference. Both test MAPE series lagged five years mimic

VXO, indicating each is overfitting to the training period. Take the period of 1997 to 1992 which has a large spike in VXO. Both models' test MAPE explodes as VXO is actually falling, indicating that the models predict much higher VXO than observed. From 1996 to 2000, training VXO is elevated and test error one year ahead declines as VXO remains elevated from 2000 to 2004. That each test MAPE series closely follows the constant model MAPE series, in which one year ahead predicted VXO is the average VXO over the five years in the training set, leads me to conclude that segmenting the training and test period based upon a time cutoff, such as before and after 1996, respectively, is a sub-optimal approach that results in overfitting.

I perform the same exercise, except rolling backwards through the same period. The first training period is 2005.01-2009.12, and the first test period is 2004.01-2004.12.



The same phenomenon is evident. With this style of training, the model clearly overfits. SVR generally performs better than the rolling mean model, but barely so.

11.12 Option implied and Realized Variance, 1990.01-2021.12

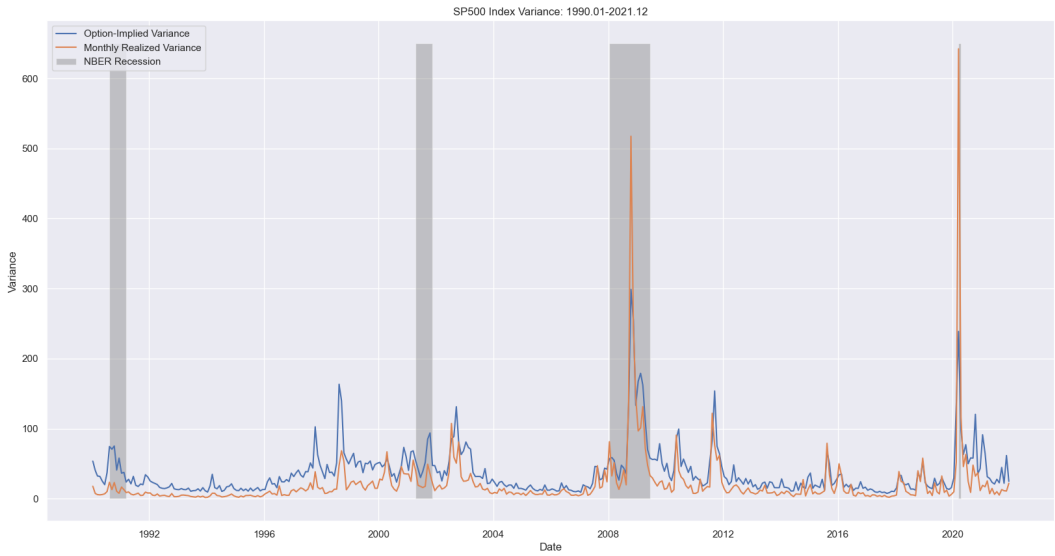


Figure 8: Option implied and Realized Variance, 1990.01-2021.12

During the Gulf War of 1991, the Peso Crisis of 1994, the Asian Tigers Crisis of 1997, and the Russian Default and LTCM blow-up of 1998, the variance risk premium was positive. Around the terrorist attacks of 9/11 and a set of accounting scandals (Enron, WorldCom), it was briefly negative. In September 2008 during the Great Financial Crisis and in March 2021 at the onset of Covid, the variance risk premium was deeply negative. For extreme events in which realized volatility spikes, the variance risk premium is negative.

11.13 Regression Measures of Test Error

Out-of-sample fit is critical to assess the model's performance. There are many metrics, including mean squared error, root mean squared error, mean percentage error, mean absolute percentage error, and the coefficient of determination, R^2. R^2 is conventionally defined as:

$$
\begin{aligned}
R^2 &\equiv 1 - \frac{\sum_{t=1}^{T} e_t^2}{\sum_{t=1}^{T}(v_t - \bar{v})^2} \\
e_t &\equiv v_t - \hat{v}_t \\
\bar{v} &\equiv \frac{1}{T}\sum_{t=1}^{T} v_t
\end{aligned}
$$

R^2 can also be expressed as a decreasing function of RMSE:

$$
\begin{aligned}
RMSE &\equiv \sqrt{\frac{1}{T}\sum_{t=1}^{T}(v_t - \hat{v}_t)^2} \\
R^2 &= 1 - \frac{\frac{1}{T}\sum_{t=1}^{T}(v_t - \hat{v}_t)^2}{\frac{1}{T}\sum_{t=1}^{T}(v_t - \bar{v})^2} \\
R^2 &= 1 - \frac{\text{RMSE}^2}{\frac{1}{T}\sum_{t=1}^{T}(v_t - \bar{v})^2}
\end{aligned}
$$

This can be computed out of sample by summing over test observations. Note, with no further assumptions, R^2 is not bounded below. An arbitrarily bad model can have high enough RMSE to make R^2 negative. In the context of OLS with a constant in the regressors, $0 \leq R^2 \leq 1$. Intuitively, such a model cannot do worse than the sample mean for predicting the response variable. Given that the sample mean of residuals equals 0, this can be rewritten to the form:

$$
\begin{aligned}
R^2 &\equiv 1 - \frac{\sum_{t=1}^{T} e_t^2}{\sum_{t=1}^{T}(v_t - \bar{v})^2} \\
R^2 &= 1 - \frac{\frac{1}{T}\sum_{t=1}^{T}(e_t - \bar{e})^2}{\frac{1}{T}\sum_{t=1}^{T}(v_t - \bar{v})^2} \\
&= 1 - \frac{\hat{\text{Var}}(e_t)}{\hat{\text{Var}}(v_t)}
\end{aligned}
$$

using the formula for the uncorrected sample variance. A negative R^2 from applying this formula to a model with no such restriction means the model does worse than simply predicting the sample mean of the dependent variable. There is no such guarantee of a fit at least as good as the sample mean for a support vector regression, hence:

$$R^2_{SVR} \neq 1 - \frac{\hat{\text{Var}}(e_t)}{\hat{\text{Var}}(v_t)}$$

In a non-linear regression, the model can perform arbitrarily worse than a constant model. When comparing model performance, I report the paper's out-of-sample R^2 using variances and conventional R^2 defined as the ratio of the sum of squared residuals to total sum of squares, along with RMSE. This permits a proper assessment of each model's performance.

I also report other prediction performance metrics, such as mean average percentage error:

$$\text{MAPE} = \frac{1}{T}\sum_{t=1}^{T}\left|\frac{v_t - \hat{v}_t}{v_t}\right|$$

to get a sense of relative error, and conventional mean squared error. A final performance metric is to regress predicted $\hat{v}_t$ on actual test v_t:

$$\hat{v}_t = \alpha + \beta v_t + u_t$$

Perfect prediction is the case for $\alpha = 0$, $\beta = 1$, and $R^2 = 1$. I compute coefficient estimates, robust standard errors, and R^2 for this regression but in the interest of brevity do not report them.

11.14 Classification Measures of Test Error

Key classification metrics are precision, recall, and F1.

$$\begin{aligned}
\text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\
\text{Precision} &= \frac{TP}{TP + FP} \\
\text{Recall} &= \frac{TP}{TP + FN} \\
\text{F1} &= \frac{2 \cdot P \cdot R}{P + R}
\end{aligned}$$

Accuracy is the number of correctly identified labels over all labels. Precision is the number of true positives over the number of labels flagged as positives. Recall is the number of true positives over the number of actual positives. For example, precision measures the percent of people diagnosed "sick" who really are sick, and recall measures the percent of people who are sick, and correctly identified by the test as such. In a medical context, you might place especially high importance on the recall score, since it's typically worse to miss a sick person than to get a false positive on a healthy person. F1 is the harmonic mean of precision and recall. It is a more useful measure than accuracy in the case of class imbalance, meaning considerably more of one class than the others. A naive classifier can guess the dominant class and obtain superficially high accuracy.

11.15 Major Historical Events: 1889.07-2009.12

Table 24: Events

1889.07-1985.12	1986.01-2009.12
Assassinations	Assassinations
1901.09: William McKinley 1914.06: Archduke Ferdinand of Austria 1963.11: John F. Kennedy 1968.04: Martin Luther King 1968.06: Robert F. Kennedy 1981.03: Ronald Reagan (Attempted)	
Natural Disasters 1900.09: Galveston Hurricane 1906.04: San Francisco Earthquake	Natural Disasters 2005.08: Hurricane Katrina
Terrorism	Terrorism
1910.10: LA Times bombing 1920.09: Wall Street bombing 1983.10: Beirut barracks bombing	1993.02: WTC bombing 1995.04: Oklahoma City bombing 1998.08: US Embassy bombings (Kenya, Tanzania) 2001.09: 9/11 2002.10: Bali bombings 2004.03: Madrid train bombings 2005.07: 7/7 London bombings 2008.11: Mumbai attacks

Table 25: Events

1889.07-1985.12	1986.01-2009.12
Financial Crises	Financial Crises
1893.05: Panic of 1893 1901.05: Panic of 1901 1907.10: Panic of 1907 1929.10-1929.11: Great Crash	1987.10: Black Monday 1994.12: Tequila Crisis (Peso devaluation) 1997.07: Asian financial crisis 1998.08: Russian default and LTCM bailout 2000.03: Dot-com bubble 2001.10–2002.12: Accounting scandals, Enron (Oct. 2001), WorldCom (June 2002), Tyco (Sept. 2002) 2008.09: Lehman bankruptcy, Fannie Mae and Freddie Mac under conservatorship 2009.03: AIG bailout
Pandemics	Pandemics
1918.02: Spanish Influenza pandemic 1957.02: H2N2 pandemic 1968.09: H3N2 pandemic	2002.11: SARS outbreak 2009.04: H1N1 Pandemic
Wars/Geopolitical Crises	Wars/Geopolitical Crises
1898.04 - 1898.98: Spanish-American War 1904.02 - 1905.09: Russo-Japanese War 1914.07 - 1918.11: WWI 1917.10 - 1917.11: Russian Revolution 1936.07 - 1939.04: Spanish Civil War 1937.07: Second Sino-Japanese war. 1939.09 - 1945.09: WWII 1948.06: Soviet Blockade of Berlin 1950.06 - 1953.07: Korean War 1956.11: Suez Crisis 1956.10 -11: Hungarian Revolution 1962.10-11: Cuban Missile Crisis 1964.08 - 1975.04: Vietnam War 1967.06: Six Day War 1968.01 - 1968.08: Prague Spring 1973.10: Yom-Kippur War 1979.11-1981.01: Iran hostage crisis 1979.12: Soviet Invasion of Afghanistan 1980.09- 1988.08: Iran-Iraq War 1982.04 - 06: Falklands War 1983.10: US invasion of Grenada	1986.04: US bombings of Libya 1989.12: US invasion of Panama 1991.01-02: Persian Gulf War 1991.12: Dissolution of USSR 1999.03-06: NATO campaign in Kosovo 2001.10: US invasion of Afghanistan 2003.03: US invasion of Iraq

11.16 Model Performance Stability with Randomized Split of Training Data

Nonparametric bootstrap methods involve repeated random sampling of data with replacement to generate an empirical sampling distribution for a statistic that may or may not have a known probability distribution. The framework is as follows (Helwig 2021). Suppose $\vec{x}$ is a random sample from a population F, for $x_i \overset{\text{iid}}{\sim} F$. The parameter $\theta = g(F)$, for arbitrary function $g(\cdot)$, is estimated with statistic $\hat{\theta} = s(\vec{x})$.

1. Sample x_i^* with replacement from $\vec{x}$ for each i=1....n.
2. Compute statistic $\hat{\theta} = s(\vec{x}^*)$ on the randomly resampled data.
3. Repeat 1-2 B times to get an empirical distribution of estimate $\hat{\theta}$.

The bootstrapped distribution of ($\hat{\theta}_1^*$,....,$\hat{\theta}_B^*$) is an empirical sampling distribution useful for estimating properties of $\hat{\theta}$ such as the standard error and a 95% confidence interval:

$$\widehat{\text{SE}}(\hat{\theta}) = \sqrt{\frac{1}{B}\sum_{b=1}^{B}\left(\hat{\theta}_b^* - \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_b^*\right)}$$

Instead of sampling individual observations, I sample partitioned C^{train} and C^{test} matrices $B = 1000$ times[46]. Each draw in step 1 is a different split of the 1986.01-2009.12 document-term matrix such that each train and test sample is comprised of different months worth of text and implied volatility data. In step 2, I fit a gradient boosting regression model and compute the out-of-sample root mean squared error and R^2. Each measure is effectively a complicated function $g(\cdot)$ of the data.

Average R^2 is 0.47 with a standard deviation of 0.12, but the 1% lows for the least favorable splits show negative R^2. Average RMSE is 6.17 with a 1.08 standard deviation. Half of bootstrapped RMSEs are over 6.03, but the 99th percentile RMSE is still lower

[46]Training a gradient boosting regression on an array with 342,155 columns is time consuming so I use $B = 1000$ randomized splits of the data to get the sampling distribution. $B \geq 10{,}000$ is recommended but I found stable results with fewer draws.

Table 26: Empirical Distribution of Test Error and R^2

	B	Mean	Std.	Min.	p1	p5	p25	p50	p75	p95	p99	Max.
RMSE	1000	6.17	1.08	3.72	4.22	4.62	5.33	6.03	6.95	8.12	8.64	10.06
R^2	1000	0.47	0.12	-0.31	0.09	0.28	0.41	0.48	0.55	0.63	0.68	0.74

Randomly split the 1986.01-2009.12 count vectorized document-term matrix with 342,155 text features B=1000 times, fit a gradient boosting regression model on each split and collect test R^2 and RMSE.

than the RMSE for forward or backward training methods. Model performance is stable across various randomized splits of the training data, giving me confidence to report 80/20 randomized split results for various methods and continue to use the randomized split approach in transfer learning. R^2 of -0.35 and 0.07 for a backwards and forwards training 80/20 temporal split are in line with the worst bootstrapped R^2, suggesting those periods are especially difficult to forecast. I plot histograms of the bootstrapped test errors:

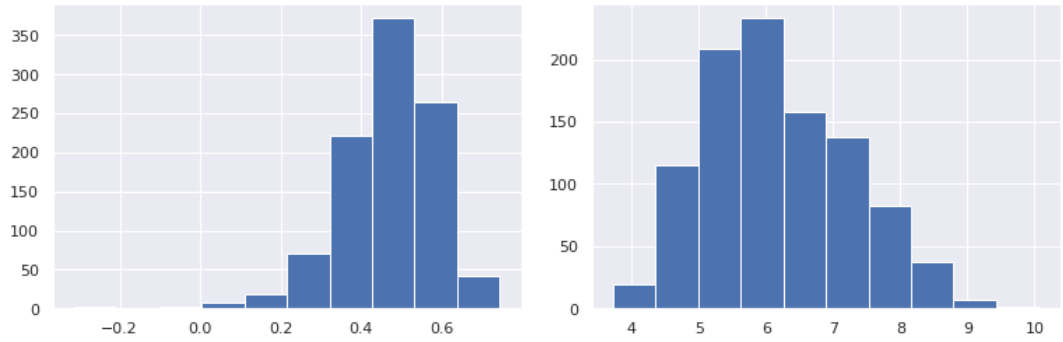


Figure 9: Bootstrapped Test R^2 (left) and RMSE (right)

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